



ACADEMIC REGULATIONS & SYLLABUS 2023 -24

**Faculty of Computer Science &
Applications**

**Smt. Chandaben Mohanbhai Patel Institute
of Computer Applications**

**Bachelor of Science in Information Technology
(B.Sc. (IT)) Programme**

(Choice Based Credit System)



**Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District:
Anand**

Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in, www.charusat.ac.in

ACADEMIC RULES

To ensure uniform system of education, duration of under graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working days' duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. Duration of Programme

Undergraduate Programme	Bachelor of Computer Applications(BCA)
Minimum	6 Semesters (3 Academic years)
Maximum	10 Semesters (5 Academic years)

3. Eligibility & Mode of Admissions

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

4. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

5. Attendance

5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

5.2 Student attendance in a course should be 80%.

6. Course Evaluation

6.1 The performance of every student in each course will be evaluated as follows:

- 6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and
- 6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

6.2 University Examination

- 6.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.
- 6.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.
- 6.2.3 Performance at Internal & University Examination will be done on the relative grading system.

7. Grading

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

Grading Scheme

Range of Marks (%)	≥80	≥75 <80	≥70 <75	≥65 <70	≥60 <65	≥55 <60	≥50 <55	<50
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

$SGPA = \sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course
 i and $i = 1$ to n , n = number of courses in the semester
 $CGPA = \sum C_i G_i / \sum C_i$
where C_i is the number of credits of course i
 G_i is the Grade Point for the course
 i and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

8. Detention Rule

A student will be promoted to next year
only if he/she has cleared all the courses of the year he/she is studying in.

9. Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

9.1.2 He should have cleared all evaluation components in every course; and

9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for the award of degree. Only latest grade will be considered.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	$CGPA \geq 7.5$ & ≤ 10
First class:	$CGPA \geq 6.0$ & < 7.5
Second Class:	$CGPA \geq 5.0$ & < 6.0
Pass Class:	$CGPA < 5.0$

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHOICE BASED CREDIT SYSTEM

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses:

Foundation Courses, Core Courses and Elective Courses.

Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core Courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

Vision, Mission, PEOs, POs and PSOs

Vision

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs) - BCA & B.Sc.(IT) Program

The graduates will:

PEO1: Be able to understand the requirement of computing problem and implement an effective solution.

PEO2: Be able to successfully take up various available career options.

PEO3: Be able to continuously learn in their preferred domains.

PEO4: Be able to acquit themselves in ethical and professional manner while demonstrating effective skills.

PROGRAM OUTCOMES (POs) - BCA & B.Sc.(IT) Program

The Graduate of Computer Science and Applications will be able to:

PO1: Computational Knowledge: Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge to provide effective solution in the area of computing.

PO2: Problem Analysis: Identify, formulate, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

PO3: Design /Development of Solutions: Design and evaluate solutions for computing problems, and design and evaluate systems, components, or processes that meet specified needs by considering several societal aspects.

PO4: Modern Tool Usage: Create, select, adapt and apply appropriate techniques, resources, and modern computing tools for any computing activities, with an understanding of the limitations.

PO5: Professional Ethics: Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

PO6: Life-long Learning: Recognise the need, and have the ability, to engage in independent learning for continual development as a computing professional.

PO7: Project management: Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO8: Communication Efficacy: Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.

PO9: Societal and Environmental Concern: Understand and assess societal and environmental issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.

PO10: Individual and Team Work: Function effectively as an individual and as a member or a leader in diverse teams and in multidisciplinary environments.

PO11: Innovation and Entrepreneurship: Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

PROGRAM SPECIFIC OUTCOMES (PSOs)- BCA and B.Sc.(IT) Program

At the end of the programme, the student should be able to

PSO1: Analyze, evaluate and formulate workable computing solutions to real world problems using computing methods.

PSO2: Apply proficiencies of communication, teamwork and management skills with spirit of entrepreneurship.

SALIENT FEATURES

- Four Years unique programme as per National Education Policy-2020.
- Exit option is available after every year (i.e 1st, 2nd and 3rd)
- Award of Degree:
 - **Certificate of Computing** (Candidate wish to Exit after the First Year)
 - **Diploma in Information Technology** (Candidate wish to Exit after the Second Year)
 - **Bachelor of Computer Science in Information Technology** (Candidate wish to Exit after the Third Year)
 - **Bachelor of Computer Science in Information Technology (Honor's)** (Candidate finishes all Four Years)
- Programme is designed based on Choice Based Credit System(CBCS).
- Right blend of fundamentals and latest technologies related courses.
- Exposure of humanities courses in every semester for holistic development of students.
- Interdisciplinary exposure in form of university elective courses.
- Unique methodologies in form of case study and project based learning for teaching-learning.
- Inclusion of environmental awareness through environment science courses.
- Coverage of emerging technologies through expert sessions, seminar and workshop.
 - Total Credits of Programme (8 Semesters): 196

**TEACHING AND
EXAMINATIONSCHEME
FOR

B.Sc.IT PROGRAMME

EFFECTIVE FROM
ACADEMIC YEAR 2023-24**

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests/ Criteria		Term work	Tests		
CA411-412 OCBSIT1001-1002	Elective-I	3	-	3	3	10	20	70	-	-	-	100
CA413	Introduction to Programming	4	3	7	7	10	20	70	15	15	70	200
CA414	Information Technology and Digital Electronics	3	-	3	3	10	20	70	-	-	-	100
CA415	Fundamentals of Database Management System	3	3	6	6	10	20	70	15	15	70	200
HS101.02 C	Communicative English	-	2	2	2	-		-	30		70	100
		13	08	21	21	400			300			700

Elective-I	
Course Code	Course Title
CA411	Fundamentals of Business Organization
CA412	Information Technology in Business
OCBSIT1001	Business to Business Marketing
OCBSIT1002	Privacy and Security In Online Social Media

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - II

Course Code	Course Title	Teaching Scheme				Examination Scheme						Total
		Contact Hours			Credit	Theory			Practical			
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA416-417 OCBSIT1003-1005	Elective-II	3	-	3	3	10	20	70	-	-	-	100
CA418	Fundamental of Commerce and business processes	3	-	3	3	10	20	70	-	-	-	100
CA419	Introduction to Object Oriented Programming	4	3	7	7	10	20	70	15	15	70	200
CA420	Fundamentals of Operating Systems	3	3	6	6	10	20	70	15	15	70	200
HS201.02C-210.02C	Liberal Arts	-	2	2	2	-		-	30		70	100
		13	08	21	21	400			300			700

Elective-II	
Course Code	Course Title
CA416	Managerial Economics
CA417	Business Models for e-Commerce
OCBSIT1003	Foundations of Cryptography
OCBSIT1004	E-Business
OCBSIT1005	Entrepreneurship Essentials

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - III

Course Code	Course Title	Teaching Scheme				Examination Scheme						Total
		Contact Hours			Credit	Theory			Practical			
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA514-515	Elective - III	3	-	3	3	10	20	70	-	-	-	100
CA516	System Analysis and Design	3	-	3	3	10	20	70	-	-	-	100
CA517	Introduction to Data Structures	4	3	7	7	10	20	70	15	15	70	200
CA518	Programming the Internet	3	3	6	6	10	20	70	15	15	70	200
HS121.02 C	Creativity, Problem Solving and Innovation	-	2	2	2	-		-	30		70	100
CL144.02 C	Environmental Sciences	-	2	2	2	-		-	30		70	100
	University Elective – I **	-	2	2	2	-	-	-	30		70	100
		13	12	25	25	400			500			900

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Introduction to Web Designing** course for others.

Elective-III	
Course Code	Course Title
CA514	Introduction to Data Warehouse
CA515	Numerical Methods

University Elective - I			
S.No.	Course Code	Course Name	Department/Faculty
1	CE281.01	Art of Programming	Engineering
2	CL281.01	Environmental Sustainability & Climate Change	Engineering
3	IT281.01	ICT Resources & Multimedia	Engineering
4	ME281.01	Engineering Drawing	Engineering
5	PH233.01	Fundamentals of Packaging	Pharmacy
6	PD260.01	Basic Laboratory Techniques	Applied Science
7	NR251.01	First Aid & Life Support	Nursing
8	PT191.01	Health Promotion & Fitness	Physiotherapy
9	BM231	Banking & Insurance	Management
10	EC281.01	Introduction to MATLAB Programming	Engineering
11	PD261	Astrophysics, Space and Cosmos-1	Applied Science
12	CL283	SDG Handprint Laboratory	Engineering
13	EE284	Python Programming	Engineering

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester – IV

Course Code	Course Title	Teaching Scheme				Examination Scheme						Total
		Contact Hours			Credit	Theory			Practical			
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA519-CA520	Elective - IV	-	3	3	3	-	-	-	10	20	70	100
CA521	Fundamentals of Visual Programming	3	3	6	6	10	20	70	15	15	70	200
CA522	Multi-paradigm programming language	-	3	3	3	-	-	-	15	15	70	100
CA523	Database Administration	4	3	7	7	10	20	70	15	15	70	200
CA524	Data Communication and Networks	4	-	4	4	10	20	70	-		-	100
HS111.02 C	Human Values and Professional Ethics	-	2	2	2	-		-	30		70	100
	University Elective – II **	-	2	2	2	-	-	-	30		70	100
		11	16	27	27	300			600			900

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Programming the Internet** course for others.

Elective-IV	
Course Code	Course Title
CA519	Introduction to Internet of Things
CA520	Fundamentals of Multimedia

University Elective - II			
S.No.	Course Code	Course Name	Department/Faculty
1	EC282.01	Prototyping Electronics with Arduino	Engineering
2	CE282.01	Web Designing	Engineering
3	CL282.01	Basics of Environmental Impact Assessment	Engineering
4	IT284	Data Visualization	Engineering
5	ME282.01	Material Science	Engineering
6	PH238.02	Cosmetics in Daily Life	Pharmacy
7	NR261.01	Life Style Diseases & Management	Nursing
8	PT192.01	Occupational Health & Ergonomics	Physiotherapy
9	BM241	Health Care Management	Management
10	CA225	Programming the Internet	Computer Applications
11	EE287	MATLAB Programming	Engineering
12	EE288	Maintenance of Household Apparatus	Engineering

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - V

Course Code	Course Title	Teaching Scheme				Examination Scheme						Total
		Contact Hours			Credit	Theory			Practical			
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA616-617	Elective-V	3	3	6	6	10	20	70	15	15	70	200
CA618	Programming with JAVA	4	3	7	7	10	20	70	15	15	70	200
CA619	Data Science	3	3	6	6	10	20	70	15	15	70	200
CA620	Object Oriented Analysis and Design	4	-	4	4	10	20	70	-	-	-	100
HS131.02 C	Communication and Soft skills	-	2	2	2				30		70	100
		14	11	25	25	400			400			800

Elective-V	
Course Code	Course Title
CA616	Advanced Internet of Things
CA617	Introduction to Game Development

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 – 24
Semester - VI

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA621	Basics of Mobile Applications	3	3	6	6	10	20	70	15	15	70	200
CA622	Software Testing	-	3	3	3	-	-	-	15	15	70	100
CA623	Mini Project	-	15	15	15	-	-	-	100		200	300
HS132.02 C	Contributor Personality Development	-	2	2	2	- -		-	30		70	100
		03	23	26	26	100			600			700

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 – 24
Semester - VII

Course Code	Course Title	Teaching Scheme				Examination Scheme							
		Contact Hours			Credit	Theory			Practical			Total	
		Theory	Practical	Total		Internal		External	Internal		External		
						Case Study	Tests		Term work	Tests			
CA624	Online MOOC Course on Current Trends of Information Technology #	-	-	-	03	-	-	-	-	-	-	-	
CA625	Minor Project	-	24	24	24	-	-	-	200		400	600	
		-	24	24	27	-			600			600	

Students can select any MOOC course of their choice related to current trends of Information Technology with at least 30 hours. 03 credits will be transferred in the result of students based on this course.

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 – 24
Semester - VIII

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA626	Dissertation	-	24	24	24	-	-	-	200	400	600	
		-	24	24	24	-			600	600		

**DETAILED SYLLABUS
FOR
B.Sc.(IT) PROGRAMME
(1ST SEMESTER)
EFFECTIVE FROM
ACADEMIC YEAR 2023 - 24**

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
 Semester - I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests/ Criteria		Term work	Tests		
CA411-412 OCBSIT1001-1002	Elective-I	3	-	3	3	10	20	70	-	-	-	100
CA413	Introduction to Programming	4	3	7	7	10	20	70	15	15	70	200
CA414	Information Technology and Digital Electronics	3	-	3	3	10	20	70	-	-	-	100
CA415	Fundamentals of Database Management System	3	3	6	6	10	20	70	15	15	70	200
HS101.02 C	Communicative English	-	2	2	2	-		-	30		70	100
		13	08	21	21	400			300			700

Elective-I	
Course Code	Course Title
CA411	Fundamentals of Business Organization
CA412	Information Technology in Business
OCBSIT1001	Business to Business Marketing
OCBSIT1002	Privacy and Security In Online Social Media

CA 411: Fundamentals of Business Organization

(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: During theory sessions, the basic concept of Business fundamentals along with its services, resources and impact in the global environment will be discussed. The core concepts of business will be covered so that students can understand the various business concepts and processes. commercial processes.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours
1	Business Fundamentals	5
2	Business Enterprises	7
3	Business Services	6
4	Emerging Trends of Business	6
5	Managing Financial Resources	5
6	Business in Global Environment	7

Total Hours (Theory): 36

Total Hours: 36

Detail Syllabus:

Unit-I: Business Fundamentals

Hours: 05

Business – Concept, nature and scope, business as a system, business objectives, business and environment interface, distinction between business, commerce and trade.

Unit-II: Business Enterprises

Hours: 07

Forms of business organization: Sole proprietorship, Joint Hindu Family Firm, partnership firm, joint stock company, types of cooperative society: Limited Liability Partnership, Choice of form of organization, multinational corporations.

Unit -III: Business Services

Hours: 06

Nature and types of Business services – Banking, Insurance, Transportation, Warehousing, communication. Meaning, role, Functions of Banking, Insurance, Postal and Telecom services, Warehousing, Transport, Hospitality and Tourism Services.

Unit -IV: Emerging Trends of Business

Hours: 06

E – business – Meaning, Scope and benefits, Resource required for successful E – Business implementation, on – line transactions, payment mechanism, Security and safety of business transactions, Outsourcing – Concept, need and scope.

Unit-V: Managing Financial Resources

Hours: 05

Functions of money, financial institutions, role of financial manager, understanding security markets, career in finance.

Unit - VI: Business in Global Environment

Hours: 07

Globalization of business, threats and opportunities in global business, global business environment, trade controls, reducing International trade barriers, career in international business.

Core Books:

1. Koontz: Principles Of Management (Ascent Series) Paperback: Tata McGraw Hill Education, 2004.
2. Y.K. Bhushan: Fundamentals of Business Organization, Sultan Chand Publisher, 2018.

Reference Books:

1. Dr. Shveta Klara, Dr. Neha Singhal: Business Organization and Management, Scholar Tech Press, 2020.
2. Karen M. Collins, Jacqueline Shemko: Exploring Business, Pearson Education, 2008.
3. Stephen J. Skripak : Fundamentals of Business : 3rd Edition, Virginia Tech Libraries, 2018.

Web References:

1. <https://vtechworks.lib.vt.edu/bitstream/handle/10919/70961/Chapter%204%20Business%20in%20a%20Global%20Environment.pdf> [Business in Global Environment]
2. <https://www.managementstudyguide.com/financial-management.htm> [Managing Financial Resources]

3. <https://leverageedu.com/blog/emerging-modes-of-business/> [Emerging Trends of Business]
4. <https://open.umn.edu/opentextbooks/textbooks/319> [ebook for reference]
5. <http://thuviensobvu.edu.vn/bitstream/TVDHBRVT/15850/1/An-Introduction-to-Business-and-Business-Planning.pdf> [Introduction to Business and Business Planning]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to understand business fundamentals
CO2 :	Able to understand different types of business proprietorship.
CO3 :	Able to get the knowledge of functionality of various online business services..
CO4 :	Able to understand the basics to manage finance for business.
CO5 :	Learned the concepts and the scope of international business enterprise.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Business Fundamentals	√				
2	Business Enterprises		√			
3	Business Services			√		
4	Emerging Trends of Business			√		
5	Managing Financial Resources				√	
6	Business in Global Environment					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	3	-	-	-	-	-	-	1
CO2	-	-	-	-	-	3	-	-	-	-	2	-	2
CO3	-	2	2	2	-	3	2	-	-	2	2	-	2
CO4	-	-	-	-	-	3	-	-	-	-	-	-	2
CO5	-	-	-	-	-	3	-	-	1	1	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA412: Information Technology in Business

(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: During theory lectures emerging trends of information technology in Business will be introduced to students. Emphasis will be given on acquisition, storage and processing of data to generate meaningful information. Students will be made familiar with applications related to information technology. Emerging trends and societal impacts of information technology will be discussed to students.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Theory Hours
1	Foundation of Information Technology	05
2	Internet Applications	05
3	Internet Service Trends in Business	06
4	E- Business and Data Analytics	07
5	Modern Business Management	07
6	Emerging Trends and Societal Impact of IT	06

Total Hours (Theory): 36

Total Hours: 36

Detail Syllabus:

Unit I: Foundation of Information Technology

Hours 05

Introduction to Information Technology, Acquisition of various types of data- Introduction to Number and Textual Data, Image Data, Audio and Video Data Numbers and Textual Data, Data Storage and Organization

Unit II: Internet Applications

Hours 05

Internet: Introduction, Internet basics, Internet protocols, Internet addressing, Browser WWW, E-mail, telnet, FTP, application, benefits and limitation of internet, electronic conferencing, and teleconferencing.

Unit III: Internet Service Trends in Business

Hours 06

Fundamentals of Cloud Computing, Types of Cloud based Services, Varieties of Cloud Computing, Types of Cloud Computing, Advantages and Disadvantage of Cloud Computing, Cloud Computing Providers, Application of Cloud Computing in Business Area. E-commerce, E-Marketing, E-Branding, E-Advertising.

Unit IV: E-Business and Data Analytics

Hours 07

E-Business: Introduction, Components, Difference between E-Commerce and E-Business ; Types of Data – Qualitative and Quantitative, Data Analytics basics, Types of Analytics, Common technologies used in data analytics, Tools for Data Analytics

Unit V: Modern Business Management

Hours 07

- E-SCM: – Introduction, Supply Chain Management, E- Supply Chain Management, Component of Modern E-SCM Major Trends in E-SCM, Example of E-SCM, Architecture of E-Supply Chain Models
- E-CRM: – Customer Relationship Management Concept, E-CRM Solution Advantages of E-CRM Solutions, Advantages of E-CRM, E-CRM Capabilities, Example of E-CRM, E-CRM Framework

Unit VI: Emerging Trends and Societal Impact of IT

Hours 06

Internet of Things, Virtual and Augmented Reality, Industry 4.0, Image Processing; Introduction to Societal Impact, Social Use of World Wide Web, Privacy, Security and Integration of Information, Disaster Recovery, Intellectual Property Right, Career in IT

Core Books:

1. V Rajaraman : Introduction to Information Technology, 2nd Edition, PHI Learning Private Limited, 2013.
2. P.T. Joseph, S.J.: E-Commerce An Indian Perspective, 5th Edition, PHI Learning Private Limited, 2015.
3. R. Kalakota & M. Robinson, e-Business 2.0, 2nd Edition, Pearson Publisher, 2012.
4. Thomas Erl, Z. Mahmood and R. Puttini: Cloud Computing Concepts, Technology & Architecture, Pearson Publication, 2014

Reference Books:

1. Syed Muhammad Fahad Akhtar, Big Data Architect's Handbook, Packt Publishing Ltd, 2018.
2. Kamlesh K. Bajaj, Debjani Nag: E-Commerce, The Cutting Edge of Business, 2nd Edition, McGraw-Hill Education, 2014.

Web References:

1. <https://blog.k2datascience.com/the-basics-of-data-analytics-77e5cc7ea741> [For Data Analytics]
2. <https://www.investopedia.com/terms/c/cloud-computing.asp> [For Cloud Computing]
3. <https://www.javatpoint.com/cloud-computing-applications> [For Cloud Computing]
4. <https://www.geeksforgeeks.org/difference-between-e-commerce-and-e-business/> [For E-business vs E-Commerce]
5. <https://www.temok.com/blog/what-is-e-business/> [For E- business]
6. <https://files.eric.ed.gov/fulltext/ED536788.pdf> [For Data Analytics]
7. <https://vdocument.in/introduction-to-e-scm.html> [For E-SCM]
8. <http://nek.istanbul.edu.tr:4444/ekos/TEZ/42370.pdf> [For E-CRM]
9. https://wps.prenhall.com/wps/media/objects/10704/10961611/Online_Appendix_B.pdf [For E-CRM]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Understand the basics of Information Technology.
CO2 :	Aware about Internet based Applications for any business context.
CO3 :	Understand trends of Internet services in Business.
CO4 :	Understands basics of Data Analytics and E-business concepts.
CO5 :	Understand modern business management concepts, emerging trends and societal Impact of Information Technology.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Foundation of Information Technology	√				
2	Internet Applications		√	√		
3	Internet Service Trends in Business			√	√	
4	E- Business and Data Analytics				√	
5	Modern Business Management					√
6	Emerging Trends and Societal Impact of IT					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1	1	-	-	2	-	-	-	-	-	2	2
CO2	3	1	2	1	-	3	-	-	-	-	2	2	2
CO3	3	1	2	1	-	3	-	-	-	-	2	2	2
CO4	3	1	2	1	-	3	-	-	-	-	2	2	2
CO5	3	1	2	1	2	2	-	-	2	-	-	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA413: Introduction to Programming

(200 Marks)

Contact Hours: 07

Pre-requisite: None

Methodology & Pedagogy: The emphasis during theory courses will be on theoretical notions like flow charts and algorithms. There will be examples of real-world problems that will be solved with the help of computers. Lectures will cover the structure of the C programming language as well as the many concepts of control statements, looping and arrays. Students will be needed to develop computer programmes in "C" in order to tackle moderately sized real-world problems during practical sessions.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Programming Basics	8	36
2	Introduction to C	9	
3	Expressions, Control Statements and Loops	10	
4	Arrays and Strings	8	
5	Functions	6	
6	Introduction to Pointers	7	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detail Syllabus:

Unit I: Programming Basics

Hours 08

Introduction to algorithm, Key features of algorithm, Introduction to flow charts, Significance of flow chart, Advantages and limitation of flow chart, Introduction to programming languages, Introduction to editor, compiler, and translator.

Unit II: Introduction to C

Hours 09

Introduction, characteristics of C, Structure of C Program, writing first C program, Files used in C program, Compiling and Executing C program; Basic data types in C, User define data types, Variables in C, Declaring and initializing variables in C, Constants, Input / Output statements in C; Operators arithmetic, relational, logical, assignment, increment- decrement, conditional, Bitwise and special, comma operator, size-of operator, operator precedence chart.

Unit III: Expressions and Control Statements

Hours 10

Arithmetic expressions, evaluation of expressions, type conversions in expressions, operator precedence and associativity, mathematical functions. Type conversion and casting; Introduction to decision control statements, Conditional and branching statements, loop, nested loop, Break, continue and go to statement.

Unit IV: Array and Strings

Hours 08

Arrays: One-dimensional, two-dimensional, Handling of Character Strings: Declaring and initializing string variables, reading string from terminal, writing string to screen, string handling functions, table of strings.

Unit V: Functions

Hours 06

Need for user-defined functions, the form of c function, return values and their types, calling a function, category of functions, handling of non-integer functions, nesting of Functions, recursion, functions with arrays

Unit VI: Introduction to Pointers

Hours 07

Understanding the Computer's Memory, Introduction to Pointers, Declaring Pointer Variables, Pointer Expressions and Pointer Arithmetic, Null Pointers, Generic Pointers, passing arguments to Function using Pointers, Pointers and Arrays, Passing an Array to a Function, Difference between Array name and Pointer, Pointers and Strings, Array of Pointers

Core Books:

1. ReemaThareja: Computer Fundamentals and Programming in C, 2nd Edition, Oxford University press, 2012
2. PradipDey and Manas Ghosh. "Programming in C", Second Edition, Oxford University Press.
3. E. Balagurusamy: Programming In C, 6th Edition, 2012

Reference Books:

1. Brian W. Kerighan and Dennis M. Ritchie : The C Programming Language, Prentice Hall.
2. Pradip Dey, Manas Ghosh: Programming in C, 4th Edition, Oxford University press, 2012.
3. Herbert Schildt: The complete reference C, Fourth edition, Tata McGraw Hill, 2000
4. Yashwant Kanetkar: Let us C, 13th Edition, BPB publication, 2013.

Web References:

1. <https://cprogrampracticals.blogspot.com/p/basic-concepts.html> [For basic c programing practical]
2. <https://www.programtopia.net/c-programming/docs/operators-expressions> [For operators and expressions in C]
3. <https://www.javatpoint.com/functions-in-c> [For functions]
4. <https://www.programiz.com/c-programming> [For basic c programing practical]
5. <https://www.javatpoint.com/c-pointers> [pointers in c]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be able to solve problems using algorithms and flowcharts.
CO2 :	Learn about the structure of C program and able to develop simple programs using the C Programming language with data types, variables and constants.
CO3 :	Gain the knowledge of different conditional and iterative statements and expressions.
CO4 :	Arrays of numbers and characters as well as manipulating with them.
CO5 :	Be able to understand user define function. Also Understand the concept of Dynamic memory management. Learn Programing-using Array with Pointer and Structure.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Programming Basics	√				
2	Introduction to C		√			
3	Control Statements and Expressions			√		
4	Arrays and Strings		√	√	√	
5	Introduction to Functions		√	√		√
6	Pointers		√			√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA414: Information Technology and Digital Electronics

(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: Fundamentals of Information technology and Digital electronics are taught in theory. Students may have clear understanding regarding the concepts of project development of Information Technology using emerging trends that can be beneficial to society. Concepts of digital electronics and techniques will be learned.

Outline of the course:

Unit No.	Title of the Unit	Theory
1	Information Technology Fundamentals, Applications and Impacts	4
2	Acquisition of Textual and Multimedia Data	5
3	CPU Fundamentals and I/O Devices Interface	6
4	Digital System and Binary Numbers	6
5	Introduction to Boolean Algebra and Logic Gates Fundamentals	7
6	Digital Logic Circuit Design	8

Total Hours (Theory): 36

Total Hours: 36

Detailed Syllabus

Unit - I: Information Technology Fundamentals, Applications and Impacts Hours: 04

Introduction to data and information, Types of Data, Simple model of computer, Applications of Information Technology in different fields (Business, Electronic Commerce, Internet Applications), Social uses of WWW, Security and Integrity of Information, Disaster Recovery, Intellectual property rights.

Unit - II: Acquisition of Textual and Multimedia Data

Hours: 05

Introduction to input units, Representation of numeric data and characters in computers, Acquisition and storage of textual data and images, Format of pictures, Image compression methods, Basics of audio signals, Acquiring and storing audio signal, Format of audio signals, audio compression methods, Basics of video signals, Acquiring and storing video signal, Format of video signals, video compression methods

Unit - III: I CPU Fundamentals and I/O Devices Interface

Hours: 06

Various input and output devices, Storage Cells, Storage Types, Memory, Structure of Central Processing Unit, Execution Mechanism of Central Processing Unit, Interface of Memory and I/O devices with CPU, Addressing Modes.

Unit - IV: Digital System and Binary Numbers

Hours: 06

Major components of digital computer, Introduction to various number systems (Decimal, Binary, Octal, Hexadecimal, BCD), Conversion of number system, Binary operations, signed binary numbers, 1's complement and 2's complement.

Unit - V: Introduction to Boolean Algebra and Logic Gates Fundamentals

Hours: 07

Fundamental concepts, Boolean Algebra Rules and Laws, Boolean expressions simplification, Canonical form and standard form, D-Morgan's Theorem, Introduction to logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR), Implementation of Boolean expressions using logic gates, Sum of product and Product of Sum forms, Universal gates, Karnaugh Map method for two, three, four variables, K-map with don't care conditions, Tabulation method.

Unit - VI: Digital Logic Circuit Design

Hours: 08

Introduction to combinational and sequential circuits, Design procedures, Adders, Subtractors, Binary adder and Subtractors, Decoder and encoders, Multiplexer and demultiplexer, Concepts of Flip-flops, Design of counters and registers, Shift registers.

Core Books:

1. M. Morris Mano: Digital Logic and Computer Design, Third Edition, Pearson Education, 2016.
2. V. Rajaraman: Introduction to Information Technology, Third Edition, PHI Learning, 2018.

Reference Books:

1. Andrew S. Tanenbaum: Structured Computer Organization, Fourth Edition, Pearson Education, 2005.
2. Abert Paul Malvino and Jerald A. Brown: Digital Computer Electronics, Third Edition, Tata McGraw Hill Publishing, 2008.
3. Thomas C. Bartee: Digital Computer Fundamentals, Sixth Edition, Tata McGraw Hill Publishing, 2012.

Web References:

1. https://www.tutorialspoint.com/digital_circuits/index.htm [Digital Circuits Tutorial]
2. <https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/> [Digital Electronics and Logic Design Tutorials]
3. <https://studymaterialz.in/digital-logic-and-computer-design-by-morris-mano/> [Digital Logic and Computer Design book]
4. <https://cupola.gettysburg.edu/oer/1/> [Digital Circuit Projects]
5. https://www.tutorialspoint.com/electronic_measuring_instruments/electronic_measuring_instruments_data_acquisition_systems.htm [Acquisition of Textual and Multimedia Data]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand IT Fundamentals and concepts of the acquisition of various data.
CO2 :	To learn memory design concepts and to study computer Arithmetic Logic Unit and Control Unit.
CO3 :	To understand the basic concepts of digital system.
CO4 :	To study logic gates for digital circuit designing.
CO5 :	Learn to design various combinational circuits and sequential circuits.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Information Technology Fundamentals, Applications and Impacts	√				
2	Acquisition of Textual and Multimedia Data	√				
3	CPU Fundamentals and I/O Devices Interface		√			
4	Digital System and Binary Numbers			√		
5	Introduction to Boolean Algebra and Logic Gates Fundamentals				√	
6	Digital Logic Circuit Design					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	2	1	1	-	3	-	-	2	1	1	1	2
CO2	2	2	2	3	-	3	2	-	2	2	2	2	2
CO3	3	3	3	3	-	3	2	-	1	2	2	3	2
CO4	3	3	3	3	-	3	2	-	-	2	2	3	2
CO5	3	3	3	3	-	3	2	-	-	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA415: Fundamentals of Database Management System

(200 Marks)

Contact Hours: 06

Pre-requisite: Basic knowledge of working with computer

Methodology & Pedagogy: During the theory lectures, concepts for data storage will be discussed. Also DBMS and RDBMS concepts will be discussed in depth. The database applications used in the real world will be discussed with necessary examples. During the laboratory hours' students will implement the concepts that are discussed during lecture.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Database System	03	36
2	Data Modeling using Entity Relationship Model	06	
3	Relational Data Modeling	06	
4	Database Design Methodology	06	
5	SQL, Schema Definition, Constraints, Queries	07	
6	Advanced Query Processing	08	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total Hours: 72

Detail Syllabus

Unit I: Introduction to Database System

Hours 03

Basic Concepts: data, **information**, **file**, database, database systems, database management system Purpose and advantages of Database management system (over file systems), **Applications of DBMS** Various data models – ER Model and Relational Model, Three level architecture, Structure of DBMS, Database actors and workers.

Unit II: Data Modeling using Entity Relationship Model

Hours 06

Entity Types, Entity Sets, Attributes, Keys, Relationship Types, Relationship Sets, Roles and Structural Constraints, Weak Entity Types, ER Diagram – Notations and fundamentals, Concept of Generalization and Specialization, Associative Entity, Conversion from ER Model to Relational Model.

Unit III: Relational Data Modeling

Hours 06

Relational Model Concepts, Relational Model Constraints, Relational Model Schemas, Update Operation, Transactions, Dealing with Constraint Violations, Relational Algebra and Calculus, Difference between ER Model and Relational Model.

Unit IV: Database Design Methodology

Hours 06

Functional Dependency and Normalization for Database – Informal Design Guidelines for Relational Schemas, Functional Dependencies, Normal Forms (1NF, 2NF, 3NF, BCNF, 4NF, 5NF), Codd Rules

Unit V: SQL, Schema Definition, Constraints, Queries

Hours 07

Introduction to SQL, why we use SQL.; Table Fundamentals: Basic Data types, Create Table Command, Viewing Data in the tables, sorting, Insert, Delete and Update Statements in SQL, Modifying the structure of tables, renaming table, truncating table, destroying table. Data Constraints and Functions: - Pseudo columns, Null values, TAB table, DUAL table Operators, Data constraints, Type of data constraints, modifying constraints, working with data dictionary and use of USER_CONSTRAINTS; Functions – introduction, merits and demerits, types of functions Numeric functions, Character functions, Date functions, Conversion functions, Aggregate functions

Unit VI: Advanced Query Processing

Hours 08

Union, Intersect and Except; Nested Queries – Introduction, Co-related Nested Queries, Set Comparison Operators Aggregate Operators, Group By clause, having clause, Order by clause Null Values – Comparison using Null Values, Logical Connectivity – AND, OR and NOT, Impact on SQL constructs, Joins (Inner Join, Outer Join, Self Join, Equi Join, Cross Join); Complex Integrity Constraints in SQL – constraints over single table, domain constraints and distinct types Creation and manipulation of database objects – indexes, views, sequences and synonym.

Core Books:

1. RamezElmasri, Shamkant B. Navathe : Fundamentals of Database Systems , 5th Edition, , Pearson Publication.
2. Ivan Bayross : SQL, PL/SQL – The programming Language Oracle.
3. Ramkrishnan, Gehrke : Database Management Systems, 3 rd Edition, McGrawHill Publication.
4. Silberschatz, Korth, Sudarshan : Database System Concepts, 5th Edition, McGraw Hill.

Reference Books:

1. C.J.Date, a Kannan, S Swaminathan : An Introduction to Database Systems, 8th Edition, Pearson Education,(Equivalent Reading).
2. Scoot Urban : Oracle 9i, PL/SQL Programming, Oracle Press.
3. S. K. Singh : Database Systems: Concepts, Design and Applications, Pearson Education
4. Peter Rob, Carlos Coronel: Database Systems: Design, Implementation and Management, 7th Edition, Cengage Learning, 2007.
5. Anjali Jivani, Amisha Shingala, Roopal and Nirav : Practice book on SQL and PL/SQL.
6. Leon and Leon : Database management Systems, Vikas Publication.

Web References:

1. <https://www.javatpoint.com/er-model-vs-relational-model> [For difference of ER and Relation model]
2. http://www.microsoftvirtualacademy.com/trainingcourses/databasefundamentals?fbid=tbZ92pOp_Tt [For overall subject]
3. http://www.ntu.edu.sg/home/ehchua/programming/sql/Relational_Database_Design.html [for relational database design]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Understand basic concepts regarding data, database systems and various data models.
CO2 :	Understand various data modeling techniques and entity relationship models.
CO3 :	Learn about relational data model and related concepts.
CO4 :	Gain insights into database design, SQL and normalization concepts.
CO5 :	Advanced query processing

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Database System	√				
2	Data Modeling using Entity Relationship Model		√			
3	Relational Data Modeling			√		
4	Database Design Methodology				√	
5	SQL, Schema Definition, Constraints, Queries				√	√
6	Advanced Query Processing					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	2	3	1	1	-	3	-	-	-	-	-	2	-
CO2	1	3	3	3	-	3	1	-	-	-	-	1	5
CO3	3	3	3	3	-	3	2	-	-	2	-	3	2
CO4	-	3	3	3	-	3	3	-	-	3	3	-	3
CO5	2	3	3	3	2	3	3	-	-	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

DETAILED SYLLABUS
FOR
B.Sc.(IT) PROGRAMME
(2ND SEMESTER)
EFFECTIVE FROM
ACADEMIC YEAR 2023 - 24

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - II

Course Code	Course Title	Teaching Scheme				Examination Scheme						Total
		Contact Hours			Credit	Theory			Practical			
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA416-417 OCBSIT1003-1005	Elective-II	3	-	3	3	10	20	70	-	-	-	100
CA418	Fundamental of Commerce and business processes	3	-	3	3	10	20	70	-	-	-	100
CA419	Introduction to Object Oriented Programming	4	3	7	7	10	20	70	15	15	70	200
CA420	Fundamentals of Operating Systems	3	3	6	6	10	20	70	15	15	70	200
FS101C	Foundation Course on Mathematics and Physics	-	2	2	2	-	-	-	50		50	100
HS201.02C-210.02C	Liberal Arts	-	2	2	2	-		-	30		70	100
		13	10	23	23	400			400			800

Elective-II	
Course Code	Course Title
CA416	Managerial Economics
CA417	Business Models for e-Commerce
OCBSIT1003	Foundations of Cryptography
OCBSIT1004	E-Business
OCBSIT1005	Entrepreneurship Essentials

CA416-Managerial Economics

(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: During theoretical lessons, real-world examples will be used to discuss economic topics such as demand, supply, production, cost function, market structure, factor pricing, and macroeconomics. Students will be offered appropriate case studies in order to provide actual exposure to theoretical economic ideas.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours
1	Basic Economic Concepts	06
2	Demand and Supply	08
3	Production, Cost and Revenue	07
4	Inflation	05
5	Market Structures & Pricing	05
6	Market Failures and Price Regulations	05

Total Hours (Theory): 36

Total Hours: 36

Detail Syllabus:

Unit - I: Basic Economic Concepts

Hours: 06

Basic of Economics, Nature of Economics, Scope of Economics, Business Economics, Difference Between Micro and Macro Economics, Laws of Economics, Business Cycle, Inflation, Needs, wants.

Unit - II: Demand and Supply

Hours: 08

Demand and supply, Types of Demand, Determinants of Demand and supply, Law of Demand and supply Demand and supply Schedule, Demand Function.

Unit - III: Production, Cost and Revenue:

Hours: 07

Production function, law of variable proportion and laws of returns to scale, different types of costs – variable cost, fixed cost, total cost, average cost, average fixed cost, average variable cost and marginal cost, Total revenue, average revenue and marginal revenue, profit function.

Unit - IV: Inflation:

Hours: 05

Meaning, and types of inflation, Causes and effect of inflation different sectors of the economy.

Unit - V: Market Structures & Pricing:

Hours: 05

Concept of market and equilibrium- characteristics of perfect competition, monopoly, monopolistic competition and oligopoly–price determinations.

Unit - VI: Market Failures and Price Regulations:

Hours: 05

Market failures and need for regulation, Regulations and market structure, Firm behavior, Price regulation.

Core Books:

1. P.L Mehta: Managerial Economics, Sultan Chand and Sons, (2014)
2. Keat Paul , K Young Philip , Erfle Steve , College Dickinson, Banerjee Sreejatha , Managerial Economics, Pearson Education; 6th ed..

Reference Books:

1. Yogesh Maheswari, Managerial Economics, Phi Learning, New Delhi, 2005
Gupta G.S.
2. Moyer & Harris. Managerial Economics, Tata McGraw-Hill, New Delhi
3. Geetika, Ghosh & Choudhury. Managerial Economics, Cengage Learning, New Delhi, 2005
4. Dominick Salvatore, Managerial Economics, adapted by Ravikesh Srivastava, Oxford University press
5. Mote V.L., Samuel Paul and G.S. Gupta, Managerial Economics Concepts and Cases, Tata McGraw Hill Publishing Company Ltd., New Delhi, 2001

Web References:

1. <https://www.geektonight.com/managerial-economics-notes/>
2. <https://www.learningclassesonline.com/2020/10/pedagogy-of-economics.html>
3. <https://mrcet.com/downloads/MBA/Managerial%20Economics.pdf>.

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to understand basic economic concepts
CO2 :	Be able to understand market
CO3 :	Student will learn basics of inflation
CO4:	Students will able to deal with balancing between cost and revenue
CO5:	Student will able to understand basics of Demand and Supply.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Basic Economic Concepts	√				
2	Demand and Supply					√
3	Production, Cost and Revenue				√	
4	Inflation			√		
5	Market Structures & Pricing:		√			
6	Market Failures and Price Regulations:		√			

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	1	2	1	-	-	-	1	1	2
CO2	-	1	1	-	1	2	2	-	-	1	3	1	2
CO3	-	1	1	-	1	2	2	-	-	-	2	1	2
CO4	1	1	1	-	1	2	2	-	-	1	2	1	2
CO5	1	1	1	-	1	2	2	-	-	1	2	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA417: Business Models for E-Commerce

(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: During theory lectures, the fundamental concepts regarding Ecommerce, Payment, E–Security, E–branding and E–Data Interchange will be discussed. Students will be required to carry out case studies using the concepts and techniques they have learnt during theory sessions.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to E-Commerce	07
2	E-Marketing	05
3	E-Security and E-Payment	08
4	Mobile Commerce	05
5	E-CRM and ERP	07
6	E-commerce Applications	04

Total Hours (Theory): 36

Total Hours: 36

Detail Syllabus:

Unit I: Introduction to E-Commerce

Hours 07

Emergence and use of Internet, origin of web and world wide web, advantages and disadvantages of Ecommerce, Features of E–Commerce, Application of E–Commerce, Electronic Commerce over the Internet, Types of E–commerce.

Unit II: E-Marketing

Hours 05

Traditional Marketing, Online Marketing, E– advertising, Internet Marketing Trends and strategies, E-branding.

Unit III: E-Security and E-Payment

Hours 08

Security on the Internet, Security Types in detail (Encryption, decryption, cryptography, Virtual Private network, Digital Signature, Secure socket Layer, Digital Certificate) E-Business risk management issues, Digital token based E-Payment System, Properties of Electronic Cash.

Unit IV: Mobile Commerce

Hours 05

Definition of M-Commerce, Features of M-Commerce, Advantages and Disadvantages of Commerce, Areas of M-Commerce Applications, Payment Systems and Models in M-Commerce.

Unit V: E-CRM and ERP

Hours 07

Customer Relationship Management, CRM capabilities and Customer life cycle, Introduction to ERP, Reasons for the growth of the ERP Market, Advantages and Disadvantages of ERP.

Unit VI: E-commerce Applications

Hours 04

Retail and Wholesale, Finance, Manufacturing, Online Booking, Online Publishing, Digital Advertising, Digital Shopping, Digital Media, Auctions, A case study should be discussed for same application.

Core Books:

1. P.T.Joseph, S.J.: E-Commerce– An Indian Perspective, 3rd Edition, PHI learning Private Limited.
2. Kamlesh K. Bajaj, Debjani Nag: E-Commerce, The Cutting Edge of Business, 2nd Edition, McGraw-Hill Education.

Reference Books:

1. Anita Rosen: The e-commerce Question and Answer Book: A Survival Guide for Business
2. Managers, 2nd Edition, Amacom.
3. Janice Reynolds: The Complete E-Commerce Book: Design, Build & Maintain a Successful
4. Web-based Business, 2nd Edition, CRC Press.
5. By Philippe Humeau&MatthieuJung : The White Book of Ecommerce Solutions, NBS Systems
6. IshitaLahiri and Sujit Kumar Ghose : Principals of Marketing and E-Commerce.

Web References:

1. <http://www.ecommercetutorial.net> [For Tutorial]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To introduce the concept of Electronic Commerce
CO2 :	To understand the concept of E-marketing
CO3 :	To understand the role of E-Security & E-Payment
CO4 :	To understand M-Commerce Payment System.
CO5 :	To learn CRM, ERP and E-commerce Applications.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to E-Commerce	√				
2	E-Marketing	√	√			
3	E-Security and E-Payment		√	√		
4	Mobile Commerce		√	√		
5	E-CRM and ERP				√	
6	E-commerce Applications					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

OCBSIT1003: Foundations of Cryptography

Description:

This is course OCBSIT1003 – **Foundations of Cryptography** is offered from SWAYAM as noc24_cs01 – Foundations of Cryptography.

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

The course provides the basic paradigm and principles of modern cryptography. The focus of this course will be on definitions and constructions of various cryptographic objects. We will try to understand what security properties are desirable in such objects, how to formally define these properties, and how to design objects that satisfy the definitions. The aim is that at the end of this course, the students are able to understand a significant portion of current cryptography research papers and standards.

Pre-requisites:

There are no pre-requisites for this course. However, it is expected that the students who are interested to take this course have had some exposure to a basic course on discrete mathematics, algorithms, or theory of computation. Having said that, I ensure that a significant effort will be made from my side to simplify the overall presentation of the course and make it easily accessible.

Industry support:

The course will be relevant for any IT related company.

Course layout:

Week 1: Course Overview, Symmetric-key Encryption, Historical Ciphers, Perfect Security and Its Limitations

Week 2: Computational Security, Semantic Security and Pseudorandom Generators (PRGs)

Week 3: Stream Ciphers, Provably-secure Instantiation of PRG, Practical Instantiation of PRG, CPA-security and Pseudo-Random Functions (PRFs).

Week 4: CPA-Secure Ciphers from PRF, Modes of Operations of Block Ciphers, Theoretical Constructions of Block Ciphers and Practical Constructions of Block Ciphers.

Week 5: DES, AES and Message Authentication Codes (MAC).

Week 6: Information-theoretic Secure MAC, Cryptographic Hash Functions, Ideal-Cipher Model, Davies-Meyer construction and Merkle-Damgård Paradigm.

Week 7: Birthday Attacks on Cryptographic Hash Functions, Applications of Hash Functions, Random Oracle Model and Authenticated Encryption.

Week 8: Generic Constructions of Authenticated Encryption Schemes, Key-exchange Problem, One-way Trapdoor Functions and Cyclic Groups.

Week 9: Discrete-Logarithm Problem, Computational Diffie-Hellman Problem, Decisional Diffie-Hellman Problem, Elliptic-Curve Based Cryptography and Public-Key Encryption.

Week 10: El Gamal Encryption Scheme, RSA Assumption, RSA Public-key Cryptosystem, KEM-DEM Paradigm and CCA-security in the Public-key Domain

Week 11: CCA-secure Public-key Hybrid Ciphers Based on Diffie-Hellman Problems and RSA-assumption, Digital Signatures, RSA Signatures and Schnorr Identification Scheme.

Week 12: Schnorr Signature, Overview of TLS/SSL, Number Theory, Interactive Protocols and Farewell.

Books and references

Text Books / Basic Material

1. Introduction to Modern Cryptography by Jonathan Katz and Yehuda Lindell.
2. Cryptography Theory and Practice by Douglas Stinson.

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$.

If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

OCBSIT1004: E-Business

Description:

This is course OCBSIT1004 – **E-Business** is offered from SWAYAM as noc24_mg16 – E-Business.

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

The Internet has changed the way companies carry out their businesses. The primary objective of this course is to introduce concepts, tools and approaches to electronic business to the post-graduate and undergraduate students. Further, the subject will help the students to develop skills to manage businesses in the digital world. The course will cover following aspects of E-Business Systems.

- Part 1: Foundations of E-Business systems
- Part 2: Infrastructure
- Part 3: Functional Areas
- Part 4: Decision Support for E-Business Systems

The course provides a balance approach including concepts from technology and management.

Pre-requisites: NIL

Industry support: All the companies.

Course layout:

Week 1: Introduction to E-Business

Week 2: Making Functional Areas E-Business Enabled : Value chain and supply chain, inter and intra organizational business processes, ERP

Week 3: Making Functional Areas E-Business Enabled : E-Procurement

Week 4: Making Functional Areas E-Business Enabled : E-marketing, E-Selling, E-Supply Chain Management.

Week 5: Technologies for E-Business: Internet and Web based system

Week 6: Technologies for E-Business: Security and payment systems

Week 7: Technologies for E-Business: Supply chain integration technologies (EDI, RFID, Sensors, IoT, GPS, GIS)

Week 8: Technologies for E-Business: Supply chain integration technologies (Web services and cloud)

Week 9: Decision Support in E-Business: Web analytics

Week 10: Decision Support in E-Business: Customer behavior modeling

Week 11: Decision Support in E-Business: Auctions

Week 12: Decision Support in E-Business: Recommender systems

BOOKS AND REFERENCES

1. Management Information Systems: Managing the Digital Firm, Laudon and Laudon, Pearson
2. Scaling for E-Business, Menasce & Almeida, PHI
3. eBusiness & eCommerce – Managing the Digital Value Chain, Meier & Stormer, Springer
4. eBook is available in springerlink.com
5. Some reference books, Internet Resources, and Research Papers

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$.

If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

OCBSIT1005: Entrepreneurship Essentials

Description:

This is course OCBSIT1005 – **Entrepreneurship Essentials** is offered from SWAYAM as noc24_ge15 – Entrepreneurship Essentials.

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

The course provides foundational knowledge on various aspects of entrepreneurial venture creation and management during its life-cycle. It has been designed to address multidisciplinary audiences. The objective of the course is to teach key issues faced by entrepreneurs and managers at different stages of the life-cycle of an enterprise and is relevant both for aspiring entrepreneurs and for decision makers in established enterprises. Topics can be classified in some major themes such as : Making a choice to create an entrepreneurial venture, current trend of technology entrepreneurship, how to start a start-up, identifying opportunities, factors driving competitive advantages, organizational structure, basic knowledge of financial statements and project report, introductory knowledge on marketing management, human resource management, & strategic management, risk analysis, legal aspect of business, how to raise fund during life-cycle of a new ventures.

Pre-requisites: All students with education up to higher secondary can take up this course.

Industry support: All start-ups, companies in emerging technology domains, companies adopting ‘corporate entrepreneurship’, professionally managed companies

Course layout:

Week 1: Introduction
Dhirubhai Ambani & Sofia
Myths & Realities about entrepreneurship
entrepreneurial qualities
Why start-ups fail?

Week 2: Mission, vision, entrepreneurial qualities – I
Mission, vision, entrepreneurial qualities – II
Value proposition
Business Model canvas
Business model generation

Week 3: Competitive advantage
Lean start-up – 1
Lean start-up – 2
Team and early recruit
Legal forms of business

Week 4: Marketing management 1
Marketing management 2
Market research –I
Market research –II
Market research –Example

Week 5: Introduction to financial statements
Profit & Loss statement
Balance sheet
Cash flow
Example – 1
Example – 2
Cost-volume-profit & Bread-Even analysis
Capital budgeting

Week 6: Business plan-I
Business plan-II
Pitching
Go-to-market strategies
Does & Don'ts.

- Week 7:** How to innovate
Design Thinking
Design-Driven Innovation, Systems thinking
Open innovation, TRIZ
How to start a start-up?
- Week 8:** Government incentives for entrepreneurship (1 lecture)
Incubation, acceleration
Funding new ventures – bootstrapping, crowd sourcing,
angel investors, VCs, debt financing (3), due diligence
Legal aspects of business (IPR, GST, Labour law)
- Week 9:** Cost, volume, profit and break-even analysis
Margin of safety and degree of operating leverage
Capital budgeting for comparing projects or opportunities
Product costing
Product pricing
- Week 10:** Funding new ventures – bootstrapping, crowd sourcing,
Angel investors, VCs, debt financing (3), and due diligence
Incubation and acceleration
Government incentives for entrepreneurship
Project cost and Financial Closure
- Week 11:** Dos & Dents in entrepreneurship
Growth Hacking
Growth Strategy
Legal aspects of business (IPR, GST, Labor law)
Negotiation skill
- Week 12:** Human Resource management in startups
Pivoting
Entrepreneurial cases
Risk assessment and analysis
Strategy management for entrepreneurial ventures
Factors driving success and failure of ventures
Concluding remarks

BOOKS AND REFERENCES

1. Effective Entrepreneurial Management: Strategy, Planning, RiskManagement, and Organization - Robert D. Hisrich • VelandRamadani, Springer (2017)
2. Entrepreneurship- Theory, Process Practice –by Kuratko &Hodgetts, Thompson South-Western Publication
3. Entrepreneurship –by Robert D. Hisrich (Edition-9)

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$.

If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

CA418: Fundamentals of Commerce and Business Processes

(100 Marks)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During theory sessions, the basic terminology along with concrete Illustrations will be discussed. The fundamentals in relation with different industries will be taught and demonstrated so that students can visualize the various business entities and commercial processes.

Outline of the Course:

Unit No.	Title of the Unit	Minimum number of hours
1	Introduction to Accounting Process and Principles	06
2	Basics of Financial Accounting	07
3	Cost Accounting	07
4	Inventory Valuation	05
5	Financial Management and Banking	06
6	Human Resource & Marketing Management	05

Total Hours (Theory): 36

Total Hours: 36

Detailed Syllabus:

Unit - I: Introduction to Accounting Process and Principles Hours: 06

Introduction to Accounting, Accounting Principles, Branches of Accounting, Accounting process of recording financial transaction, Concept of Book-keeping, Accounting system, Classification of Accounts, Rules of Debit & Credit

Unit - II: Basics of Financial Accounting Hours: 07

Accounting system, Single and Double entry system, Journal entry, Ledger and Trial balance, Gross and Net Trial Balance, Trading Account, Profit & Loss Account, Balance Sheet.

Unit - III: Cost Accounting

Hours: 07

Meaning and scope of Cost Accounting, Objectives of Cost Accounting, Difference between Financial and Cost Accounting, Components of Cost, Types of Cost, Costing and Cost Accounting, Methods of Costing, Marginal Costing, Breakeven Analysis, Make or buy decision.

Unit - IV: Inventory Valuation

Hours: 05

Introduction of inventory, Classification of Inventory, Periodical and Perpetual Inventory Systems, Inventory Valuation Techniques: FIFO, LIFO, WAC, Applications of Inventory Valuation. Methods of pricing, FIFO, LIFO, Weighted average, Treatment of returns from previous issues and shortages identified at stores verification.

Unit - V: Financial Management and Banking

Hours: 06

Meaning and functions of Financial Management, sourcing of funds and cost of funds, Time value of money, Introduction to Banking, Deposits Management, Types and Schemes, Management of Advances, Types of Management, Banking Services, Banking Products.

Unit - VI: Human Resource & Marketing Management

Hours: 05

Human Resource: HR planning in context of software industries, Recruitment, Selection, Types of Interview, Training and Development, Welfare and working conditions. Marketing Management: Selling and Marketing comparison, Core concepts of marketing, Pricing Decisions, Advertising and Sales promotion.

Core Books:

1. Dr. S. N. Maheshwari: Principles of Management Accounting, Sultan Chand Publication. Eighteenth Edition, 2021
2. M. Pande: Financial Management, Vikas Publishing House, 2015
3. V. S. P. Rao: Human Resource Management, Excel Books, 2010

Reference Books:

1. Dr. S.N. Maheshwari: Advanced Accountancy, Volume 1, Eleventh & Revised Edition, Vikas Publishing House, 2017
2. M.N. Arora: Cost Accounting, Vikas Publishing House, 2013
3. B.S.Mathur: Banking Law & Practice.
4. Paresh P Shah, Financial Management, Bizantra Publication, 2008

Web References:

1. <https://ncert.nic.in/textbook/pdf/keac101.pdf> [Introduction of Accounting]
2. <http://gbpssi.in/admin/coursepack/MBR517Lect02.pdf> [Final Accounts]
3. <https://catalogimages.wiley.com/images/db/pdf/9780471791454.excerpt.pdf> [Accounting Elements]
4. <https://wirc-icai.org/images/material/Valuation-Inventories-Accounting-Investments-&-Cash-Flow-Statement-PG.pdf> [Inventory Valuation]
5. <http://www.jiwaji.edu/pdf/ecourse/commerce/Unit-1%20INTRODUCTION%20TO%20FINANCIAL%20MANAGEMENT.pdf> [Introduction to Financial Management]
6. <http://mpbou.edu.in/slm/mba1p6.pdf> [Basic concepts of Marketing]

Course Outcomes: Upon successful completion of the course, the students will:

CO1 :	Be able to understand areas of accounting and learn Financial Accounting in detail.
CO2 :	Be familiar with Costing and understand Break-even analysis, and determine cost using different methods and techniques.
CO3 :	Be able to understand periodical and perpetual inventory and apply inventory calculation methods.
CO4 :	Be able to understand the Financial Management and Business Organization.
CO5 :	Be able to understand different types of sectors and their processes.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		Co1	Co2	Co3	Co4	Co5
1	Introduction to Accounting Process and Principles	√				
2	Basics of Financial Accounting		√			
3	Cost Accounting			√		
4	Inventory Valuation				√	
5	Financial Management and Banking					√
6	Human Resource & Marketing Management					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	2	2	3	3	-	-	-	-	3	3	2
CO2	3	2	2	-	3	2	3	-	-	2	-	2	3
CO3	2	3	3	-	3	3	-	-	-	2	2	3	2
CO4	3	3	3	3	2	3	2	-	-	-	3	3	3
CO5	3	-	2	-	3	2	-	-	-	2	2	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA419: Introduction to Object Oriented Programming (200 Marks)

Contact Hours: 07

Pre-requisite: CA413 - Introduction to Programming.

Methodology & Pedagogy: During the Theory sessions concept will be covered with real world examples to understand the application and usage of object oriented programming over procedural programming. During Practical sessions, students will implement the concepts using various assignments and case study.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	OOP Concept and Introduction to C++	07	36
2	Classes and Object	08	
3	Polymorphism	08	
4	Inheritance	09	
5	Exception Handling and Template	09	
6	Managing I/O formats and operations	07	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detail Syllabus:

Unit I: OOP Concept and Introduction to C++

Hours 07

Introduction to C++, Structure of C++ program, Fundamental of OOP, Characteristics of OOP, inline function, pass by reference, default arguments, creating namespace.

Unit II: Classes and Object

Hours 08

Classes and objects in C++, Control flow statements: if else, for loop, while loop, do while loop, switch, break and continue, Constructors: default, parameterized, copy, constructor overloading and destructor, Access specifiers, implementing and accessing class members. Working with objects: constant objects, nameless objects, live objects, arrays of objects, Introduction to functions, library and user-defined functions, parameters passing, default arguments.

Unit III: Polymorphism

Hours 08

Polymorphism: Introduction, Types of Polymorphism, Function overloading, friend functions, Operator overloading: Introduction, overloaded operators, operator keyword, operator return values, unary operator overloading, binary operators overloading, overloading with friend function, rules of overloading operators.

Unit IV: Inheritance

Hours 09

Inheritance: Introduction, derived class declaration, Types of inheritance, Inheritance and member access ability, Constructor and Destructor in derived class, Virtual Base Class. Virtual functions, Pure Virtual function, Compile time Vs Runtime, early vs. late binding.

Unit V: Exception Handling and Template

Hours 09

Introduction to Exception handling, catching class types, using multiple catch statements, exception handling options, creating custom exceptions, Template: Introduction, Class Template, Function Template, Standard Template Library.

Unit VI: Managing I/O formats and operations

Hours 07

Understand and apply concepts from various header files like iostream. Iomanip, fstream, etc. opening and closing file, reading and writing text files and binary files, performing random access on files, handle command line arguments

Core Books:

1. Herbert Schildt: C++: A Beginner's Guide, 2nd Edition Paperback, McGraw-Hill, 2003.
2. E Balagurusamy: Object Oriented Programming with C++ Paperback, McGraw Hill India, 2017.

Reference Books:

1. Yashavant Kanetkar: Let Us C++, Paperback, BPB Publications, 2020.
2. Robert Lafore: Object Oriented Programming in C++, 4th Edition, Sams Publications, 2002.
3. Bruce Eckel: Thinking in C++, Volume 1, 2nd Edition, Pearson Education, 2006.

Web References:

1. <https://www.geeksforgeeks.org/c-plus-plus/> [For Entire Syllabus]
2. <https://google.github.io/styleguide/cppguide.html> [For coding standards and concepts]
3. http://www.tutorialspoint.com/cplusplus/cpp_basic_input_output.htm [For Practicals]

Course Outcomes: Upon successful completion of the course, the students will:

CO1:	Understand the basic structure of C++ program and understand the concept of object oriented programming characteristics.
CO2:	Understand the concept of Constructor and Destructor and able to create class and objects.
CO3:	Understand the need of Polymorphism and implement static polymorphism
CO4:	Understand the purpose of Inheritance, identify relations between classes, implement inheritance and perform dynamic binding.
CO5:	Able to handle the runtime errors or exceptions in C++ and use templates. Able to read input from device, file, template, command line arguments and apply formats to input and output.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		Co1	Co2	Co3	Co4	Co5
1	OOP Concept and Introduction to C++	✓				
2	Classes and Object, Function & Access Specifier		✓			
3	Polymorphism and Types of Overloading			✓		
4	Inheritance				✓	
5	Exception Handling and Templates					✓
6	Managing I/O formats and operations					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	2	2	2	2	2	2	2	2	2	3	2
CO2	3	3	3	2	2	2	3	2	2	2	2	3	2
CO3	3	2	3	2	2	2	3	2	2	2	2	3	2
CO4	3	3	3	2	2	2	3	2	2	2	2	3	2
CO5	3	2	3	2	2	2	2	2	2	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA420: Fundamentals of Operating Systems

(200 Marks)

Contact Hours: 06

Pre-requisite: None

Methodology & Pedagogy: In theory sessions the students shall learn the basic principles of an operating systems and in practical sessions the students will implement the concepts using windows and Linux operating systems using batch files and shell scripting.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Basics of Operating Systems	5	36
2	Process Management	7	
3	Processor Management	7	
4	Memory Management	7	
5	File Management	5	
6	Device Management	5	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total Hours: 72

Detailed Syllabus:

Unit - I: Basics of Operating Systems

Hours 05

What is an operating system, Brief history of an operating system, Pyramid structure of an operating system, Operating system hardware and software, Evolution of an operating system, types of operating system, Batch system, Interactive system, Real time system, Hybrid system, Shell, compiler, assembler, interpreter?

Unit - II: Process Management

Hours 07

Concept of Process and thread, Process Control Block, States of Process, Process Models- Two State, Five State, Seven State, Deadlock, Necessary Conditions for Deadlock or Live lock, Modeling Deadlocks, Strategies for Handling Deadlocks, Starvation.

Unit - III: Processor Management

Hours 07

Process scheduler, Job scheduler, various process states, Thread control, Process Control Block (PCB), Processor scheduling policies, FCFS, SJN, SRT, RR, EDF, Multiple level queues

Unit - IV: Memory Management

Hours 07

Fixed partition, Dynamic partition, Best fit and first fit allocation method, De-allocation, Paged memory allocation, Demand paging allocation, segmented memory allocation, Virtual memory, cache memory, Associative memory, Page replacement policies.

Unit - V: File Management

Hours 05

Fundamentals of File management, File naming conventions, Fixed length and variable length records, Physical file organization, Physical storage allocation, File access methods, File compression.

Unit - IV: Device Management

Hours 05

Types of devices, dedicated device, Virtual device, shared device, I/O devices in cloud, Sequential Access Storage Media, Direct access storage media, Device handler seek strategies, FCFS, SSTF, SCAN and LOOK strategies, disk storage.

Core Books:

1. Ann McIver McHoes, Ida M Flynn: Understanding Operating System, 7th Edition, Cengage Learning, 2014
2. William Stallings: Operating Systems: Internals and Design Principles, 9th Edition, Pearson, 2018.
3. Sumitabha Das: Unix concepts & applications, 4th Edition, Tata McGraw Hill, 2010.
4. Ken Rosen, James Farber, Rachel Klee, Douglas Host, and Dick Rosinski, “ Unix: The Complete Reference”, 2nd edition, McGraw Hill Education, 2007

Reference Books:

1. Abraham Silberschatz, Greg Gagne, Peter B. Galvin: Operating System Concepts, 10th Edition, Wiley, 2018.
2. Mark G. Sobell: A Practical Guide to Linux Commands, Editors and Shell Programming, 3rd Edition, Prentice Hall, 2013.
3. K.J. George, Operating System Concepts 9th Edition, Shroff Publishers and Distributors Pvt. Ltd., 2003.

Web References:

1. <https://study.com/learn/operating-system.html> [Operating System Video Lessons]
2. <https://www.studytonight.com/operating-system/> [Operating System Tutorial]
3. <http://www.tutorialspoint.com/unix/index.htm> [Unix Tutorials]
4. <http://williamstallings.com/OS/Animation/Animations.html> Animations of Operating Systems Concepts]
5. <http://www.ics.uci.edu/~ics143/lectures.htm> [Lecture notes of OS]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand the fundamental concepts of an operating System.
CO2 :	To understand the concept of process management.
CO3 :	To understand the processor scheduling policies
CO4 :	To understand the memory allocation and de- allocation methods.
CO5 :	To study various supported device and file access technologies.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Basics of Operating Systems	√				
2	Process Management		√			
3	Processor Management			√		
4	Memory Management				√	
5	File Management					√
6	Device Management					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	1	2	2	-	2	2	-	1	1	-	3	2
CO2	3	2	2	2	-	2	2	-	1	1	-	3	2
CO3	3	2	2	2	-	2	2	-	1	1	-	3	2
CO4	3	2	2	2	-	2	2	-	1	1	-	3	2
CO5	2	2	3	3	-	2	3	-	1	1	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS201.02 C: PAINTING (100 Marks)

Contact Hours: 30 Hrs.

Pre-requisite: None

Instructional Methods and Pedagogy: Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

Evaluation: The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation: Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation: University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Outline of the course:

Unit No.	Title of the unit	Minimum number of hours
1	An Introduction to Painting	02
2	Drawing from Nature and Object	04
3	Colour Design and Colour Value	06
4	Composition and Perspective	06
5	Figure Drawing and Proportion	04
6	Sketching	04
7	Contemporary Issues in Painting	04

Practical Hours: 30 Hrs.
Total hours: 30 Hrs.

Unit – I	An Introduction to Painting	02 Hours
	• An Introduction to Painting • Principles of Composition • Medium and Techniques of Painting • History of Painting: Folk Indian Painting / Western Painting • 2D and 3D Painting	
Unit – II	Drawing from Nature and Object	04 Hours
	• Objects of Drawing: Nature and Manmade / Artificial Objects • Drawing Still / Live Objects • Drawing from Memory • Drawing from Life	
Unit - III	Light and Shade	06 Hours
	• Color Theory: Color wheel (primary/secondary, complementary), transparency/opacity, hue, value (intensity, brightness), chroma (saturation, purity) & temperature (warm/cold) • Color Contrast & Attributes: Interaction, harmony, psychology/mood, culture & expression • Media Characteristics & Surfaces: Acrylic, oil, paper, wood & canvas (primed/unprimed)	
Unit – IV	Composition and Perspective	06 Hours
	• Composition: Space, movement, balance, asymmetry, rhythm, shapes, proportion & lighting • Perspective: An approximate reproduction • Types of Perspectives: Linear Perspective, One-point Perspective, Two-point Perspective, Three-point Perspective, Four-Point Perspective	
Unit – V	Figure Drawing and Proportion	04 Hours
	• Proportions of the Human Body • Three views – Anterior (front), Lateral (side) and Posterior (back) • Fundamental Proportion – The Big Three	
Unit – VI	Sketching	04 Hours
	• Sketching and Freehand • Sketching Techniques • Sketch and Drawing Medium	

Unit VII

Contemporary Issues in Painting

**04
Hours**

- Contemporary Indian Art
- Pioneers of Contemporary Indian Art
- Contemporary Issues in Painting

Reference Books

1. A.N. Hodge (2007) *The History of Art: The Essential Guide to Painting Through the Ages* Arcturus Foulsham, London
2. David Lewis (1984) *Pencil Drawing Techniques* Watson-Guption Publications, New York
3. John C Van Dyke L H D (1895) *A Text Book of the History of Painting* Longmans Green and Co., London
4. Sarah Parks (2014) *Drawing Secrets Revealed - Basics How to Draw Anything*-North Light Books, Ohio

Web Material

1. <https://conceptartempire.com/what-are-the-fundamentals/>
2. <https://www.creativebloq.com/art/painting-techniques-artists-31619638>
3. <https://www.thesprucecrafts.com/perspective-in-paintings-2578098>
4. <https://www.thehansindia.com/posts/index/Hans/2016-05-19/Origin-of-painting-in-India/229138>

Course Outcome (COs): At the end of the course, the students will be able to:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art history, art criticism and aesthetics.
CO3	be able to recognize the elements of arts in painting.
CO4	have better cognizance and association of meaning of colors, shapes, and composition.
CO5	be able to acknowledge the principles of painting as in design and colors, concept, medium and formats.
CO6	have instantaneous painting experience about designing, lights, shades and colors and such other important aspects.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	2	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	-	-	3	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS202.02 C: PHOTOGRAPHY (100 Marks)

Contact Hours: 30 Hrs.

Pre-requisite: None

Instructional Methods and Pedagogy: Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

Evaluation: The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation: Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation: University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Photography	03
2	Camera and Operating System	05
3	Light and Shade	10
4	Composition	09
5	Contemporary Issues in Photography	03

Practical Hours: 30 Hrs.
Total hours: 30 Hrs.

Unit – I	An Introduction to Photography	03 Hours
	• Art, Design and Visualization • Basics of Photography and Various Types of Photography • Basics of Post Production • A Brief History of Photography: • Early Experiments and Later Developments	
Unit – II	Camera and Operating System	05 Hours
	• Role of Camera in the Photography • Types of Camera Pin-hole, box, folding, large and medium format cameras, single lens reflex (SLR) and twin lens reflex (TLR), miniature, subminiature and instant camera • Principal Parts of Photographic Camera Lens, Aperture, Shutters, various types and their functions, focal plane shutter and in-between the lens shutter, shutter synchronization, self-timer • Types of Lenses Single (meniscus), achromatic, symmetrical and unsymmetrical lenses, telephoto, zoom, macro, supplementary and fish-eye lenses • Different Models of Camera, their Features and Operating Systems • Camera and Size of the Image, Speed and Power of Lens	
Unit – III	Light and Shade	10 Hours
	• Reflection and refraction of light, Dispersion of light through a glass prism, lenses • Colour Filters: Different kinds, Red, yellow, green, neutral density, half filters, filter factor, colour correction filter • Photographic Light Sources: Natural source, the Sun, nature and intensity of the sunlight at different times of the day, different weather conditions • Artificial light sources: Nature, intensity of different types of light sources used in photography namely; (i) Photo flood lamp, (ii) Spot light, (iii) Halogen lamp, Barn doors and snoot, lighting stands Flash unit: Bulb flash and Electronic flash, main components, electronic flash units, studio flash, slave unit, multiple flash, computer flash, x-contact, exposure table.	
Unit – IV	Composition	09 Hours
	• Different kinds of image formations • Principal focus and focal length of the lens • Depth of field, angle of view and perspective • Perspective and composition	

Unit – V Contemporary Issues in Photography

**03
Hours**

- Rules of composition
- Present Day Photography
- Contemporary Photographers and their Contributions
- Major Issues in Contemporary Photography

Reference Books

1. Maria Short, Sri-Kartini Leet and Elisavet Kalpaxi (2020) *Context and Narrative in Photography*-Routledge, New York.
2. Kevin Los (2010) *Improve Your Photography: 50 Essential Digital Photography Tips & Techniques* Digital Photography Series, New York.

Web Material

1. <https://photographylife.com/photography-basics>
2. <https://artofvisuals.com/the-basics-of-photography-introduction-to-photography-tutorials/>
3. <https://www.photographytalk.com/beginner-photography-tips/camera-basics-for-beginners>
4. <https://www.makeuseof.com/tag/5-photography-tips-for-beginners/>
5. <https://annamcnaught.com/blog/basics-of-photography>

Course Outcome (COs): At the end of the course, the students will be able to:

CO1	Understand, appreciate and demonstrate innovative approach, beauty and acute acumen in the area of photography
CO2	Develop photography skills and become familiar with the functions and importance of the visual elements of nature and artificial objects
CO3	Become independent thinkers who will contribute inventively and critically to culture through the making of art photography
CO4	Have thorough understanding and acute sense of light and shade, composition, and presentation of a piece of an art
CO5	Experiment and Represent the cultivated sense and skills in Photography to the mass
CO6	Prepare an impressive portfolio encompassing holistic approach to art and other the areas of study

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	2	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	2	-	-	-
CO4	-	-	-	-	-	2	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	2	-	-	3	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS205.02 C: MEDIA AND GRAPHIC DESIGN (100 Marks)

Contact Hours: 30 Hrs.

Pre-requisite: None

Instructional Methods and Pedagogy: Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

Evaluation: The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation: Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation: University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Media and Graphic Design	03
2	Layout and Design	07
3	Form and Space	06
4	Computer Graphics	04
5	Fonts	04
6	Basic Print Media	03
7	Contemporary Issues in Graphic Design	03

Practical Hours: 30 Hrs.
Total hours: 30 Hrs.

Unit – I	An Introduction to Media and Graphic Design	03 Hours
	• Creating Art, Art in Context and Art as Inquiry • History of Graphic Design • Constructional, Representational, and Simplification Drawing	07 Hours
Unit – II	Layout and Design	
	• Layout, Design and Aesthetics • Elements of Design • Principles of Design: Harmony, Balance, Rhythm, Perspective, Emphasis, Orientation, Repetition and Proportion • Impact/function of Design • Indigenous design practices • Role of design in the changing social scenario	06 Hours
Unit – III	Form and Space	
	• Types of Forms: Man-made, Nature • Types of Space: Negative and Positive • Composition of Form and Space to create Layout Exploring Creativity	04 Hours
Unit – IV	Computer Graphics	
	• An Introduction to Graphic Software • Flash, Coreldraw, Illustrator and Photoshop • Pre-press Process	04 Hours
Unit – V	Fonts	
	• Construction of Type • Anatomy of Type • Visual Language • Creating Logo and Symbol	03 Hours
Unit – VI	Basic Print Media	
	• An Introduction to Press and its Development Phases • Types of Press • Types of Printing Technologies • Post-press Processes	03 Hours
Unit – VII	Contemporary Issues in Graphic Design	
	• Present Day Graphic Designs • Contemporary Designers and their Contribution • Major Contemporary Issues in Graphic Design	

Reference Books

1. Ellen Lupton and Jennifer Cole Philips (2008) *Graphic Design: The New Basics*, Princeton Architectural Press, New York
2. Ryan Hembree (2011) *The Complete Graphic Designer: A Guide to Understanding Graphics and Visual Communication* Rockport Publishers, Beverly

Web Material

1. <https://99designs.com/blog/tips/graphic-design-basics/>
2. <https://edu.gcfglobal.org/en/beginning-graphic-design/fundamentals-of-design/1/>
3. <https://ultragraphicsmt.com/10-basic-concepts-to-improve-your-graphic-design/>
4. <https://www.jcsocialmedia.com/graphic-design/>

Course Outcome (COs): At the end of the course, the students will be able to:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art and designs, art criticism and aesthetics.
CO3	be able to recognize the elements of arts in graphic design.
CO4	have better cognizance and association with the meaning of designs, shapes, colors, print and medium.
CO5	be able to design graphics using computer softwares like Photoshop, CorelDraw, and Illustrator.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	3		-	2	-	-	-	-
CO4	-	-	3	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	3	-	-	-	-	2	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS209.02 C DRAMATICS

(100 Marks)

Contact Hours: 30 Hrs.

Pre-requisite: None

Instructional Methods and Pedagogy: Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

Evaluation: The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation: Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation: University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Drama	06
2	History of Drama and Contemporary Theatre	06
3	Theatre Design and Techniques	06
4	Technicalities of Stage Performance	08
5	Contemporary Trends in Drama	04

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

Detailed Syllabus:

Unit – I	Introduction to Drama	06 Hours
	• Introduction to performing arts • Drama - An art, a socializing activity, & a way of learning • Form of Drama • Elements of Drama • Types of Drama	
Unit – II	History of Drama and Contemporary Theatre	06 Hours
	• Important world dramatists & drama—from Greek to modern • Evolution of contemporary theatre in the context of developments in Indian theatre • Major Movements in Drama	
Unit – III	Theatre Design and Techniques	06 Hours
	• Theatre Architecture • Stage craft: Set, light, costume, make up, sound, props • Theatre techniques: from selection of script to final performance	
Unit – IV	Technicalities of Stage Performance	08 Hours
	• Selection of plot and character • Improvisation • Movement • Voice, Speech, Imagination • Character Development • Scene Enactment	
Unit – V	Contemporary Trends in Drama	04 Hours
	• New Tendencies in theatre • Drama and Society • Using drama for Social Change and Education	

Reference Books

1. Bridget Panet (2009) Essential Acting: A Practical Handbook for Actors, Teachers and Directors Routledge, New York.
2. Tom Bancroft (2012) Character Mentor: Learn by Example to Use Expressions, Poses, and Staging to Bring Your Characters to Life Focal Press, New York

Web Material

1. <https://entertainism.com/elements-of-drama>
2. <https://www.rcboe.org/cms/lib/GA01903614/Centricity/Domain/5069/the%20elements%20of%20drama.pdf>
3. <https://marrzipandrama.co.nz/the-6-elements-of-drama/>
4. <https://www.backstage.com/magazine/article/theater-trends-broadway-2019-66726/>

Course Outcome (COs): At the end of the course, the students will be able to:

CO1	be aware about the concept of performing art and its nuances.
CO2	display a working knowledge of historic of drama, its development and current trends in dramatics.
CO3	demonstrate skills in the technical/design preparation and execution of a theatre performance.
CO4	demonstrate the ability to work collaboratively.
CO5	develop essential transferable skills in various relevant areas of the theatre.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	2	-	-	2	-	-	-	-
CO3	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	3	-	-	-
CO5	-	-	-	-	-	-	-	-	2	-	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS210.02C - CONTEMPORARY DANCE

(100 Marks)

Contact Hours: 30 Hrs.

Pre-requisite: None

Instructional Methods and Pedagogy: Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

Evaluation: The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation: Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation: University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to dance	04
2	Types of Dance	06
3	Basic Elements of Dance	04
4	Technical Skills in Professional Contemporary	06
5	Contemporary Trends in Dance	10

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

Detailed Syllabus:

Unit – I	Introduction to dance	04 Hours
	• Dance as a Performing Art • Dance as a Medium of Expression • History and Development of Dance	
Unit – II	Types of Dance	06 Hours
	• Western dance and classical dance • Salsa, rumba, hip hop, tap dance, belly dance, etc. • Indian Classical Dance forms: Odissi, Bharatanatyama, Kathak, Kathakali, Kuchipudi etc. • Other Regional dance forms in India	
Unit – III	Basic Elements of Dance	04 Hours
	• Movements of different parts of a body for Expression • Concepts of: Nritya, Laya and Taal	
Unit – IV	Technical Skills in Professional Contemporary Dance	06 Hours
	• Dance technique: alignment, balance, co-ordination, flexibility and control • Expressive / presentation skills: Dynamic energy, physical engagement with the given material and stage, etc. • Skills and processes of rehearsal and production: • physical energy, stamina and athleticism	
Unit – V	Contemporary Trends in Dance	10 Hours
	• Prevalent trends and techniques in contemporary dance • Future trends in contemporary dance form • On Stage Performance	

Reference Books

1. Helene Scheff, Marty Sprague and Susan Mc Greeve. (1939) *Experiencing dance from student to dance artist*. Human kinetics, New York
2. Robin Grove, Catherine J. Stevens, Shirley McKenzie (2005) *Thinking in Four Dimensions: Creativity and Cognition in Contemporary Dance*. Melbourne university press, Melbourne

Web Material

1. <https://www.liveabout.com/what-is-contemporary-dance-1007423#:~:text=Contemporary%20dance%20is%20a%20style,body%20through%20of%20dance%20movements.>
2. <https://dancemagazine.com.au/2014/01/whats-contemporary-dance-days/>
3. <https://www.frontiersin.org/articles/10.3389/fpsyg.2019.00071/full>
4. <https://www.thehindu.com/features/friday-review/dance/what-is-contemporary-dance/article5096022.ece>

Course Outcome (COs): At the end of the course, the students will be able to:

CO1	be able to develop ability to express through the form of dance.
CO2	have enhanced aesthetic sensitivity.
CO3	have improved concentration, mental alertness, quick reflex action, and physical agility.
CO4	be able to express a natural way human feelings and expressions by creating harmony.
CO5	be able to deliver contemporary dance performance.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	3	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	-	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**DETAILED SYLLABUS
FOR
B.Sc.(IT) PROGRAMME
(3RD SEMESTER)
EFFECTIVE FROM
ACADEMIC YEAR 2023 - 24**

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - III

Course Code	Course Title	Teaching Scheme				Examination Scheme						Total
		Contact Hours			Credit	Theory			Practical			
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA514-515	Elective - III	3	-	3	3	10	20	70	-	-	-	100
CA516	System Analysis and Design	3	-	3	3	10	20	70	-	-	-	100
CA517	Introduction to Data Structures	4	3	7	7	10	20	70	15	15	70	200
CA518	Programming the Internet	3	3	6	6	10	20	70	15	15	70	200
HS121.02 C	Creativity, Problem Solving and Innovation	-	2	2	2	-		-	30		70	100
CL144.02 C	Environmental Sciences	-	2	2	2	-		-	30		70	100
	University Elective – I **	-	2	2	2	-	-	-	30		70	100
		13	12	25	25	400			500			900

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer

Introduction to Web Designing course for others.

Elective-III	
Course Code	Course Title
CA514	Introduction to Data Warehouse
CA515	Numerical Methods

University Elective - I			
S.No.	Course Code	Course Name	Department/Faculty
1	CE281.01	Art of Programming	Engineering
2	CL281.01	Environmental Sustainability & Climate Change	Engineering
3	IT281.01	ICT Resources & Multimedia	Engineering
4	ME281.01	Engineering Drawing	Engineering
5	PH233.01	Fundamentals of Packaging	Pharmacy
6	PD260.01	Basic Laboratory Techniques	Applied Science
7	NR251.01	First Aid & Life Support	Nursing
8	PT191.01	Health Promotion & Fitness	Physiotherapy
9	BM231	Banking & Insurance	Management
10	EC281.01	Introduction to MATLAB Programming	Engineering
11	PD261	Astrophysics, Space and Cosmos-1	Applied Science
12	CL283	SDG Handprint Laboratory	Engineering
13	EE284	Python Programming	Engineering

CA514: Introduction to Data Warehouse

(100 Marks)

Contact Hours: 36

Pre-requisite: Fundamentals of Database Management System

Methodology & Pedagogy: During theory lectures emphasis will be given on understanding fundamental concepts of data warehouse and its architectures along with the different applications. Also, the emphasizes given on using OLAP and OLTP tools for understanding the real time applications of data warehouse technologies.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Theory Hours
1	Introduction Data Warehouse	6
2	Data Warehouse Architecture	8
3	OLAP & OLTP. OLAP Tools	6
4	ETL & ELT Process in Data Warehouse	7
5	Key Understanding of Data Warehouse Terminologies and Tools	5
6	Applications of Data Warehouse and Business Intelligence	4

Total Hours (Theory): 36

Total Hours:36

Detail Syllabus:

Unit I: Introduction to Data Warehouse

Hours 06

Introduction to Data Warehouse, Types of Data Warehouse, Difference between Database and Data Warehouse, Why Data Warehouse, Scope, Redundancy of data and End User of Data Warehouse. Developing Strategy of Data Warehouse.

Unit II: Data Warehouse Architecture

Hours 08

Data Warehouse Design Consideration and Dimension Modeling. Evolving Data Warehouse Architecture, Data Warehouse Architecture — Learn Architecture, Concepts and Components, Components of Data Warehouse. Data Warehouse Design Methodologies.

Unit III: OLAP & OLTP. OLAP Tools

Hours 06

Data Integration, Concept, Process and Techniques, types of OLAP and differences among them. Introduction to various OLAP tools.

Unit IV: ETL & ELT Process in Data Warehouse

Hours 07

What is ETL, what is ELT, Key Differences between ETL & ELT, ETL & ELT Tools, Concept of Automation Testing in Context of ETL & ELT.

Unit V: Key Understanding of Data Warehouse Terminologies and Tools

Hours 05

Overview and understanding of integral terminologies of Data Warehouse like Fact Table, Dimensional Table, Subject Oriented, Time Variant, Extraction, Transformation, Data Lake, Data Mart, Task Relevant Data, Loading, Various types of Query, Homogeneous Database, Heterogeneous Database, Functional Dependency, Transitive Dependency. Overview of various tools available for Data Warehouse.

Unit VI: Applications of Data Warehouse and Business Intelligence

Hours 04

Explore broad categories of Data Warehouse applications, Financial Services, Banking Services, Consumer Goods, Retail Sector, and Controlled Manufacturing.; Implications of Data Warehouse in the field of Data Mining & Business Intelligence.

Core Books:

1. William H. Inmon: Building the Data Warehouse, 4th edition, Wiley publication, 2005
2. Paulraj Ponniah: Data Warehousing: Fundamentals for IT Professionals, 2nd edition, 2010

Reference Books:

1. Nenad Jukic: Database Systems: Introduction to Databases and Data Warehouses 1st Edition, Prospect Press, 2017

Web References:

1. ETL vs ELT: Must Know Difference Between ETL and ELT (guru99.com) [For ETL & ELT]
2. DATA WAREHOUSING UNIT 1,2,3,4,5,6 (mu.ac.in) [Data Warehouse Tutorial]
3. Data Warehouse Tutorial for Beginners: Learn Basic Concepts (guru99.com) [Data Warehouse terminologies]
4. Data Lineage Done Right - Data Lineage Tool - MANTA (getmanta.com) [Data Warehouse Tools]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Have a deeper understanding of database systems and their underlying theory to be able to improve the decision-making process.
CO2 :	Understand the technology of data warehousing.
CO3 :	be able to develop applications of higher order database systems
CO4 :	Be able to understand and maintain complex and heterogeneous data.
CO5 :	Clear understanding of implementation of huge amount of static data

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction Data Warehouse	√				
2	Data Warehouse Architecture		√			
3	OLAP & OLTP. OLAP Tools			√		
4	ETL & ELT Process in Data Warehouse			√		
5	Key Understanding of Data Warehouse Terminologies and Tools				√	
6	Applications of Data Warehouse and Business Intelligence					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA515-Numerical Methods

(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: During theory lectures illustrations of certain real world problems requiring numerical computations inference will be explained. Numerical methods will be introduced as tools for solving such problems also be emphasized.

Outline of the course:

Unit No.	Title of the Unit	Theory
1	Introduction to Numerical Methods	4
2	Interpolation and Regression	6
3	Numerical Integration and Differentiation	6
4	Solution of Linear Equation	7
5	Solution of Non-Linear Equation	7
6	Solution of Ordinary Differential Equations	6

Total Hours (Theory): 36

Total Hours: 36

Detailed Syllabus:

Unit - I: Numerical Methods

Hours: 04

Introduction to Numerical Methods, Various methods such as Bisection, False-Position and Newton-Raphson methods

Unit - II: Interpolation and Regression

Hours: 06

What is an interpolation, Difference tables, Newton forward and backward Interpolation formula, Lagrange's formula, Newton's Divided Difference Formula, introduction of Regression, Interpolation versus Regression, Linear and Non Linear Regression?

Unit - III: Numerical Integration and Differentiation

Hours: 06

Numerical Integration, Trapezoidal rule, Simpson's $1/3$ & $3/8$ rules, Differentiating Continuous function (Two point and Three-point formula), Tabulated function and its maxima and minima.

Unit - IV: Solution of Linear Equation

Hours: 07

Solution of Linear Equation, Gauss-Jordan Elimination methods, Gauss - Seidal and Gauss Jacobi Interactive methods.

Unit - V: Solution of Non-Linear Equation

Hours: 07

Errors in numerical calculations, sources of errors, solving non-linear equations using trial and error method, Half interval method and convergence, Newton's method and convergence.

Unit - VI: D Solution of Ordinary Differential Equations

Hours: 06

Solution of Ordinary Differential Equations, Taylor Series and Euler Methods, Runge-Kutta Methods,

Core Books:

1. D. N. Datta: Computer Oriented Numerical Methods, Vikas Pub. House, 2012.
2. S.S. Sastry : Introductory methods of Numerical Analysis, Prentice Hall of India, 4th edition 2006

Reference Books:

1. V. Rajaraman Computer Oriented Numerical Methods, 3rd Edition, Prentice-Hall of India Pvt.Ltd, 3rd edition, 2016
2. M. Ray, Har Swarup Sharma, Sanjay Chaudhary: Mathematical Statistics, Ram Prasad and sons, 11th edition
3. B S Grewal: Numerical Methods In Engineering & Science (With Programs In C, C++ And MATLAB), Khanna Publishers, 10th edition Edition, 2014

Web References:

1. [https://ece.uwaterloo.ca/~dwharder/NumericalAnalysis/10RootFinding/bisection/example s.html](https://ece.uwaterloo.ca/~dwharder/NumericalAnalysis/10RootFinding/bisection/example%20s.html) [Bisection Method]
2. [https://ece.uwaterloo.ca/~dwharder/NumericalAnalysis/10RootFinding/falseposition/exa mples.html](https://ece.uwaterloo.ca/~dwharder/NumericalAnalysis/10RootFinding/falseposition/examples.html) [False Position Method]
3. <https://paws.kettering.edu/~ktebeest/math305/simp38b.pdf>[Simpson's-1/3 & Simpson's- 3/8]
4. <https://hamrocsit.com/note/csc207/solution-of-nonlinear-equations/> [solutions of linear equations]
5. <https://sites.google.com/site/nmmat202/file-cabinet> [Lecture notes of Numerical methods]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to use Numerical methods in computer technology.
CO2 :	To study Interpolation and Regression.
CO3 :	To Learn the concepts of Numerical Integration and Numerical Differentiation
CO4 :	To understand linear and non-linear equations.
CO5 :	To learn techniques to solve ordinary differential equations

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Numerical Methods	√				
2	Interpolation and Regression		√			
3	Numerical Integration and Differentiation			√		
4	Solution of Linear Equation				√	
5	Solution of Non-Linear Equation				√	
6	Solution of Ordinary Differential Equations					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	1	-	2	-	-	-	2	-	2	1
CO2	3	3	2	1	-	2	-	-	-	2	-	2	1
CO3	3	3	2	1	-	2	-	-	-	2	-	2	1
CO4	3	3	2	1	-	2	-	-	-	2	-	2	1
CO5	3	3	2	1	-	2	-	-	-	2	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA516: System Analysis and Design (100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: During theory lectures emphasis will be given on understanding how Information Systems are designed and constructed and how major modifications are made to existing systems, analyze business needs for information and develop an appropriate strategy to solve a particular business problem, understand the issues associated with physical design, utilize some of the tools used by systems analysts in the design of Information Systems, a significant portion of the course will deal with computer aided system engineering (CASE) tools and techniques in the context of a case study.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours
1.	Introduction of information systems	05
2.	System analysis and Information gathering	07
3.	System Requirement specification	07
4.	Data flow diagrams and Process Specification	06
5.	System Design and Quality assurance	06
6.	Control- Audit, Maintenance and conversion of information systems	05

Total Hours (Theory): 36

Total Hours: 36

Detail Syllabus:

Unit I: Introduction of information systems

Hours 05

Concept of system, components of system, characteristics of system, computer-based system, information system categories, management structure. Introduction to information, types of information, quality of information, information.

Unit II: System analysis and Information gathering

Hours 06

Problem definition, feasibility study, Analysis – design and development of system.; Introduction to system analyst, role of system analyst, attributes of system analyst.; Strategy to Gather Information, Information Sources, Methods of Searching for Information, Interviewing Technique, Questionnaires, Other Methods of Information Search.

Unit III: System Requirement specification

Hours 07

System Requirements Specification, Data Requirements, steps in Systems Analysis, Modularizing Requirements Specifications, deciding on Project Goals, Examining Alternative Solutions, Evaluating Proposed Solution, Cost-benefit Analysis, Feasibility report;

Unit IV: Data flow diagrams and Process Specification

Hours 07

Symbols Used in DFDs, describing a System with a DFD, Levelling of DFDs, Levelling Rules, Logical and Physical DFDs, Case Tools to Draw DFD, Process Specification Methods, Structured English, Decision Table Decision Tree;

Unit V: System Design and Quality assurance

Hours 06

Process and system design stages, logical and physical design, Data Input, Coding techniques, validating input data, interactive data input, objective of output designing, design of screens and output reports. Introduction to quality assurance, levels of quality assurance, the testing process with test data: Program testing, Link testing, Full system testing with live data.

Unit VI: Control- Audit, Maintenance and conversion of information systems

Hours 05

Control in Information Systems, The Objectives of Control, Control Techniques, Audit of Information Systems, Security of Information Systems, designing reliable and maintainable systems, design of software, software design and documentation tools, managing quality assurance, user training, training methods and conversion;

Core Books:

1. James A Senn: Analysis and Design of Information System, McGraw Hill International, 2003.
2. V. Rajaraman: Analysis and Design of Information Systems, Prentice-Hall of India Private Limited, 2003.
3. Kendall and Kendall: Systems analysis and Design, 5th Edition, Prentice-Hall of India Private Limited, 2003.

Reference Books:

1. Jeffrey L. Whitten, Lonnie D. Bentely and Kevin C. Dittman : Systems Analysis and Design Methods, Tata McGraw Hill Publishing Co. Ltd., 2001.
2. Tuthill and Leavy : Knowledge Based Systems : Mangers Perspectives : Tab professional and Reference Books, 1991.

Web References:

1. <http://lecture-notes-forstudents.blogspot.in/2010/04/system-analysis-and-design-sad.html> [For referring SDLC]
2. <http://www.slideshare.net/aroravinay/1-introduction-to-ado-infosys>[For Introduction of System analysis and design]
3. <http://bcastuff.blogspot.in/p/sad-notes.html#.U6uW-ZzWIeo>[For brief notes on System analysis]
4. <http://www.eis.mdx.ac.uk/staffpages/geetha/bis2030/DFD.html>[For DFD notes and examples]
5. <http://faculty.washington.edu/ytan/is460/notes/LN11.pdf>[For DFD notes and examples]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Understand the basics of system, type of information systems, system analyst-meaning, role and need, portfolio development of the system, the methods or ways to review and select the project.
CO2 :	Understand the concept of SDLC, the fact finding techniques, detailed study of modern methods for collecting the requirements of system to be developed, Decision and procedure documentation and a brief introduction to the SRS document.
CO3 :	Learn the concept of structured analysis, detail study of DFD and its types and the study of data dictionary.
CO4 :	Gain understanding of prototypes, its uses and tools for development, clarification of the misconception about the prototype and categories of tools including the use of automated tools like case tools.
CO5 :	Be able to learn objective of collecting data, methods to collect data for designing the input and output interface, the concept and objective of database design
CO6 :	Study the designing of reliable and maintainable systems, documentation tool, quality assurance management and need for user training.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		Co1	Co2	Co3	Co4	Co5	Co6
1	Introduction of information systems	√					
2	System analysis and Information gathering		√				
3	System Requirement specification			√			
4	Data flow diagrams and Process Specification				√		
5	System Design and Quality assurance					√	
6	Control- Audit, Maintenance and conversion of information systems						√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	2	2	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	3	3	2	2	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	2
CO5	1	3	3	2	2	2	3	2	2	2	2	3	2
CO6	1	2	2	2	3	3	3	2	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA517: Introduction to Data Structures

(200 Marks)

Contact Hours: 07

Pre-requisite: Introduction to Programming and Introduction to Object Oriented Programming.

Methodology & Pedagogy: During theory lectures emphasis will be given on understanding fundamental concepts of data structures and algorithms and how to use and choose the most efficient data structure from a variety of available ones for different kind of applications. During practical session emphasizes given on designing, analyzing and implementation of all kind of data structure for solving computational problems using high level programming language.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Data Structures and Algorithms	05	36
2	Linear Data Structures Stack and Queue	08	
3	Dynamic Data Structure	09	
4	Trees Data Structure	09	
5	Graph Data Structure	09	
6	Sorting and Searching	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detail Syllabus:

Unit I: Introduction to Data Structures and Algorithms

Hours 05

Data type, atomic and composite data, Abstract Data type, Data object, Relationship among Data, Data Structure, characteristics of a data structure, need for data structure, Execution time cases, Basic terminology, Types of Data Structures, Introduction to Algorithms, Greedy Algorithms, Divide and Conquer Algorithm, Analysis of Algorithms, space and time complexity of an algorithm.

Unit II: Linear Data Structures Stack and Queue

Hours 08

Linear Data Structures using sequential organization, Concepts of Ordered list, Representation of Stacks using sequential organization, Applications of Stack, Recursion, Application of Recursion, Concepts of Queues, Realization of Queues using sequential organization, Dequeue, Priority Queue, Applications of Queue.

Unit III: Dynamic Data Structure

Hours 09

Introduction to DDS, linked list, Comparison of sequential and linked organizations, Dynamic Memory Management, singly linked list, doubly linked list, Circular linked list, Applications of linked list.

Unit IV: Trees Data Structure

Hours 09

Basic Terminologies, Definition and concepts, Representation of Binary Tree, Operations on Binary Tree and algorithms, Types of Binary Tree, BTree, B+Tree Indexing, AVL tree, heap tree.

Unit V: Graph Data Structure

Hours 09

Graph Terminology, Representation of Graph, Operations on Graph, Applications of Graph Structure, minimum spanning tree, shortest path algorithms.

Unit VI: Sorting and Searching

Hours 08

Sorting Notations and Concepts, Sorting Techniques, Sequential Searching, Binary Searching, Interpolation Search, Search Trees, Hash Table Methods.

Core Books:

1. Michael T. Goodrich, Roberto Tamassia, David Mount: Data Structures and Algorithms in C++, Wiley; Second edition -2016.
2. R. S. Salaria: Data Structures & Algorithms Using C++, Khanna Book Publishing - 2015.
3. R. Singh, I. Singh: Data Structures: An Algorithmic Approach with C++, Khanna Book Publishing - 2014.
4. D. Samanta, Classic Data Structures, 2nd Edition, PHI Publications, 2002.

Reference Books:

1. Yashvant Kanethkar : Data structures through C++, 2nd Edition, BPB Publications, 2003.
2. Varsha h. Patil, Data Structures Using C++, 1st Edition, Oxford University Press, 2012.

3. Yedidyah Langsam, Moshe J. Augenstein, Aaron M. Tenenbaum : Data structures using C and C++ , 2nd Edition, Eastern economy edition, Pearson Education, 2015.
4. Mark Allen Weiss: Data Structures and Algorithm Analysis in C++, 3rd Edition, Pearson Education, 2017.
5. Reema Thareja: Data Structures using C, 1st Edition, Oxford University Press, 2012.

Web References:

1. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/ds_ToC.html [For materials of Data Structures and Algorithms]
2. <http://www.cs.usfca.edu/~galles/visualization/> [For Data Structures Visualization]
3. <https://www.cs.auckland.ac.nz/~jmor159/PLDS210/mst.html> [For Graph Data Structures]
4. <http://interactivepython.org/runestone/static/pythonds/Trees/trees.html> [For Tree Data Structures]
5. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/niemann/s_man.pdf [For Sorting and Searching Cook book]
6. <http://www.cs.princeton.edu/~rs/AlgsDS07/10Hashing.pdf> [For Hashing Materials]
7. <http://nptel.ac.in/video.php?subjectId=106102064> [For Data Structures videos]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Create and follow suitable data structures for fixing computing problems
CO2 :	Understand basic ADTs such as stacks, queues, and trees
CO3 :	Explore and implement concepts of dynamic memory allocation and usage of it through linear data structures like link list
CO4 :	Students will familiar with non-linear data structure like tree and graph
CO5 :	Apply Algorithm for solving problems like sorting, searching

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Data Structures and Algorithms	√				
2	Linear Data Structures Stack and Queue		√			
3	Dynamic Data Structure			√		
4	Trees Data Structure		√		√	
5	Graph Data Structure				√	
6	Sorting and Searching					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	-	2	2	2	2	-	-	-	2	3
CO2	2	3	2	2	-	2	-	-	1	-	1	2	3
CO3	3	3	2	2	-	2	-	-	1	-	1	3	3
CO4	2	3	3	2	-	2	-	-	1	-	1	2	2
CO5	3	3	3	2	-	2	-	-	1	-	1	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA518: Programming the Internet

(Marks 200)

Contact Hours: 07

Pre- requisite: None.

Methodology & Pedagogy: During theory sessions, topics related to web designing technologies will be covered with suitable examples. During Practical sessions, students will be required to design and develop entire web sites using several web designing technologies and editors.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1.	Basics in Web Design	06	36
2.	Web Page Designing using HTML5	10	
3.	Introduction to Cascading Style Sheets	10	
4.	JavaScript Programming I	08	
5.	JavaScript Programming II	07	
6.	Web Publishing & Search Engine Optimization	07	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detailed Syllabus:

Unit I: Basics of Web Design

Hours: 06

Overview of Internet and WWW, Basic elements of the Internet, Internet services, Internet Browsers and Servers, Introduction to WWW, URL, webpage, web site, web servers, web browser, Web Application, Client and Server side scripting languages, types of websites, Web Design and Development, Internet Addressing: standard Internet Address, Domain Name Server (DNS).

Unit II: Webpage Development using HTML5

Hours: 10

HTML Fundamentals: HTML & its relevant history, Anatomy of an HTML Tag, Basic HTML Document Structure, working with Text, working with Lists, Tables and Frames, working with Hyperlinks, Images and Multimedia, Working with Forms and Controls. Advanced Elements in HTML5: Semantic Elements, New Input Type Elements, Multimedia Tags

Unit II: Introduction to Cascading Style Sheets

Hours: 10

Introduction to Cascading Style Sheet (CSS), basic syntax and structure, CSS selectors, Ways of specifying style, CSS Properties, CSS Styling (Background, Text, Fonts, Lists, Tables, Links) , CSS Box Model , CSS Navigation Bar; CSS3: - CSS Rounded Corners, Box & Text Shadow , Gradients, Background Images, Transitions, Transforms, and Animations, CSS Layout

Unit IV: JavaScript Programming I

Hours: 08

Introduction to Client Side Scripting, Basics of Java Script, Java Script Statements, Comments, variables, Operators and Expressions, Conditions statement, Functions, Dialog boxes.

Unit V: JavaScript Programming II

Hours: 07

The Java Document Object Model (DOM): JavaScript Document Object Model hierarchy – Create find and manipulate HTML Element using Objects and methods. Form validation, Applying Style using JavaScript. Creating New window, Accessing & manipulating, History of HTML Pages; Forms: Form object, built in objects, User defined objects, Cookies, Java Script Window; DHTML: Introduction to DHTML, DHTML CSS, DHTML Java Script, DHTML HTML DOM, DHTML Events.

Unit VI: Web Publishing & Search Engine Optimization

Hours: 07

FTP Management: Understanding FTP, setting up FTP Server, Uploading and downloading FTP contents; Search Engine Optimization: Using Good Keywords for Keyword Search, Creating the Title Tag and Meta Tags for Website, Optimizing Website for Search Engine - Keyword Density and Content, Submitting Site to the Search Engines, Increase Rankings by Building Link Popularity

Core Books:

1. Matthew MacDonald: HTML5 The Missing Manual, O'Reilly Media, August 2011.
2. Peter Gasston: The Book of CSS3, A Developer's Guide to the Future of Web Design, No Starch Press, April 2011.
3. Richard York: Beginning CSS Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2005.

Reference Books:

1. Ivan Bayros : Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP, fourth revised edition , BPB Publication.
2. David Mc Farland : CSS The Missing Manual, O'Reilly, 2006.
3. Julie C. Meloni : HTML, CSS and JavaScript All in One, Pearson.

Web References:

1. <http://www.tutorialspoint.com/html5> [For notes on HTML5 tags]
2. <http://www.w3schools.com/html/> [For HTML5, CSS and JavaScript notes and examples]
3. <https://in.godaddy.com/help/dreamweaver-cs6-publish-your-website-7811> [Publish your website using Dreamweaver]
4. <http://fullbooksfreedownload.blogspot.in/2016/02/html-css-javascript-web-publishing-in.html> [Book :- HTML, CSS & JavaScript Web Publishing in One Hour a Day, Sams Teach Yourself, 7th Edition PDF]

Course Outcomes: Upon successful completion of the course, the students will:

CO1 :	Get familiar with basics of the Web Designing & Publishing.
CO2 :	Gain deep understanding of the implementation of HTML tags and able to create HTML static web pages.
CO3:	Able to use basic properties of CSS and CSS3 in html file
CO4 :	Able to create dynamic web page using JavaScript
CO5 :	Able to understand the strategies of search engine optimization

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		Co1	Co2	Co3	Co4	Co5
1	Basics in Web Design	√				
2	Web Page Designing using HTML5		√			
3	Introduction to Cascading Style Sheets			√		
4	JavaScript Programming I				√	
5	JavaScript Programming II				√	
6	Web Publishing & Search Engine Optimization					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	1	-	2	-	-	-	2	-	1	1
CO2	3	2	2	1	-	2	-	-	-	2	-	2	1
CO3	3	3	3	1	-	2	-	-	-	2	-	2	1
CO4	3	3	2	1	-	2	-	-	-	2	-	2	1
CO5	3	3	2	2	-	2	-	-	-	2	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS121.02 C: CREATIVITY, PROBLEM SOLVING AND INNOVATION

(100 Marks)

Contact Hours: 03

Pre-requisite: Creative Problem Solving (<https://www.coursera.org/learn/creative-problem-solving>)

Outline of the Course:

Unit No.	Title of the Unit	Minimum number of hours
1.	Introduction to Creativity, Problem Solving and Innovation	06
2.	Questioning, Learning and Visualization	06
3.	Creative Thinking and Problem Solving	06
4.	Logic, Language and Reasoning	06
5.	Contemporary Issues and Practices in Creativity and Problem Solving	06

Total Hours (Practical): 30

Total Hours: 30

Detailed Syllabus:

Unit - I	Introduction to Creativity, Problem Solving and Innovation Definitions of Creativity and Innovation, Need for Problem Solving and Innovation, Scope of Creativity in various Domains, Types and Styles of Thinking, Strategies to develop Creativity, Problem Solving and Innovation skills	Hours: 06
Unit - II	Questioning, Learning and Visualization Strategy and Methods of Questioning, Asking the Right Questions, Strategy of Learning and its Importance, Sources and Methods of Learning, Purpose and Value of Creativity Education in real life, Visualization strategies - Making thoughts Visible, Mind Mapping and Visualizing Thinking	Hours: 06
Unit - III	Creative Thinking and Problem Solving Creative Thinking and its need, Strategy of Thinking Fluency, Generating all Possibilities, SCAMPER Technique, Divergent Vs Convergent Thinking, Lateral Vs Vertical Thinking, Fusion of Ideas for Problem Solving, Applying strategies for Problem Solving	Hours: 06
Unit - IV	Logic, Language and Reasoning Basic Concepts of Logic, Statement Vs Sentence, Premises Vs Conclusion, Concept of an Argument, Functions of Language: Informative, Expressive and Directive, Inductive Vs Deductive Reasoning, Critical Thinking & Creativity, Moral Reasoning	Hours: 06
Unit - V	Contemporary Issues and Practices in Creativity and Problem Solving Cognitive Research Trust Thinking for Creatively Solving Problems, Case Study on Contemporary Issues and Practices in Creativity and Problem Solving	Hours: 06

Text book:

1. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity, Jossy-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genius, Ten Speed Press 2001

Reference book:

1. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
2. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
3. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
4. Tony Buzan, How to Mind Map, Thorsons Publication 2002
5. Scott Berkun, The Myths of Innovation, Expanded and revised edition, Berkun Publication 2010
6. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
7. Ira Flatow, The all Laughed, Harper Publication 1992
8. Paul Sloane, Des MacHale & M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

Additional Readings

1. Keith Sawyer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
2. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
3. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
4. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
5. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
6. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
7. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
8. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994
9. Scott G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
10. Donald J Treffinger, Scott G Isaksen, K Brainstead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
11. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
12. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
13. Howard Gardner, Creating minds, Basic Books Publication 1993
14. Mihaly Csikszentmihalyi, Creativity—Flow and Psychology of Discovery and Invention, Harper Publication 1996
15. Martin Gardner, W. H., Aha! Insight, Freeman Publication 1978
16. Paul Sloane, Test Your Lateral Thinking IQ, Sterling Publication 1994

Web material:

1. Internet Search based May TED talks and other sources for videos, slide shares, problems, etc.

Course Outcome (COs): At the end of the course, the students will be able to

CO1	Demonstrate creativity in their day to day activities and academic output.
CO2	Solve personal, social and professional problems with a positive and an objective mindset.
CO3	Think creatively and work towards problem solving in a strategic way.
CO4	Initiate new and innovative practices in their chosen field of profession.
CO5	Give logical ideas, opinions, and solutions to problems.
CO6	Think critically over the situation and drawing conclusion.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	2
CO2	-	-	3	-	-	-	-	-	3	-	-	-	2
CO3	-	-	-	-	-	-	-	-	3	-	-	-	2
CO4	-	-	-	-	3	-	-	-	3	-	-	-	2
CO5	-	-	-	-	-	-	-	-	3	-	-	-	2
CO6	-	-	-	-	-	-	-	-	3	-	-	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

DETAILED SYLLABUS
FOR
B.Sc.(IT) PROGRAMME
(4TH SEMESTER)
EFFECTIVE FROM
ACADEMIC YEAR 2023 - 24

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester – IV

Course Code	Course Title	Teaching Scheme				Examination Scheme						Total
		Contact Hours			Credit	Theory			Practical			
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA519-CA520	Elective - IV	-	3	3	3	-	-	-	10	20	70	100
CA521	Fundamentals of Visual Programming	3	3	6	6	10	20	70	15	15	70	200
CA522	Multi-paradigm programming language	-	3	3	3	-	-	-	15	15	70	100
CA523	Database Administration	4	3	7	7	10	20	70	15	15	70	200
CA524	Data Communication and Networks	4	-	4	4	10	20	70	-		-	100
HS111.02 C	Human Values and Professional Ethics	-	2	2	2	-		-	30		70	100
	University Elective – II **	-	2	2	2	-	-	-	30		70	100
		11	16	27	27	300			600			900

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Programming the Internet** course for others.

Elective-IV	
Course Code	Course Title
CA519	Introduction to Internet of Things
CA520	Fundamentals of Multimedia

University Elective - II			
S.No.	Course Code	Course Name	Department/Faculty
1	EC282.01	Prototyping Electronics with Arduino	Engineering
2	CE282.01	Web Designing	Engineering
3	CL282.01	Basics of Environmental Impact Assessment	Engineering
4	IT282.01	Internet Technology & Web Design	Engineering
5	ME282.01	Material Science	Engineering
6	PH238.01	Cosmetics in Daily Life	Pharmacy
7	NR261.01	Life Style Diseases & Management	Nursing
8	PT192.01	Occupational Health & Ergonomics	Physiotherapy
9	BM241	Health Care Management	Management
10	PD262	Astrophysics, Space and Cosmos-2	Applied Science
11	EE287	MATLAB Programming	Engineering
12	EE288	Maintenance of Household Apparatus	Engineering

CA519: Introduction to Internet of Things (100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: In order to achieve the course objectives, students will be introduced to the basics of embedded system, Internet of Things (IoT) based system and their structures. Students will be introduced to various microcontrollers, analog and digital electronics components and sensors. Introduction to microcontroller based Internet of Things (IoT) system design with specifications, development and implementation will be explained. Students will also be introduced to basic programming of an embedded system and IOT system.

Outline of the course:

Week No	Practical	Description
1 - 2	Introduction to Internet of Things	Introduction - Overview of Internet of Things (IoT), The building blocks of IoT, Various technologies making up IoT ecosystem, IoT levels, IoT design methodology, The Physical Design and Logical Design of IoT, Functional blocks of IoT and Communication Models, Development Tools used in IoT.
3 - 4	Things, Clouds and Connections	Working of Controlled Systems, Real-time systems with feedback loop, different type of modes using wired and wireless methodology, the process flow of an IoT application. Introduction to various cloud platform used in IoT system such as Thingworx, Microsoft Azure, Google cloud, Thingspeak etc..
5 - 6	Programming Internet of Things System	Arduino / node MCU controller for IOT Programming, Embedded C data types, variables, constants and operators, programming using control

		statements, loops, string and array, Arduino IDE.
7 - 8	Interfacing Microcontroller with Hardware	Interfacing Arduino / nodeMCU with various input / output devices like LEDs, Push button switch, Toggle switch, 7 segment display, LCD interfacing, Relay, keypad matrix and other electronics devices.
9 - 10	Sensors and Actuators for Internet of Things System	Interfacing with various sensors like temperature sensor, PIR sensor and ultrasonic sensor.
11 - 12	Designing Mobile Application and Webpage for Internet of Things System	Basic concepts of mobile application development, designing webpage for IoT user, Connecting App / webpage with cloud and controller as per Internet of Things system levels.

Total Hours (Practical): 36

Total Hours:36

Core Books:

1. Arshdeep Bahga, Vijay Madisetti: Internet of Things: A Hands-On Approach, VPT Publication, 2014.
2. Michael Margolis, Brian Jepson, Nicholas Robert Weldin: Arduino Cookbook: Recipes to Begin, Expand, and Enhance Your Projects , O'Reilly Media Publisher, 3rd Edition, Kindle Edition, 2020.

Reference Books:

1. Muhammad Azhar Iqbal, Enabling the Internet of Things, John Wiley & Sons,2020.
2. Massimo Banzi, Michael Shiloh: Getting Started with Arduino: The Open Source Electronics Prototyping Platform ,Make Communication, 3rd Edition, Kindle Edition, 2014.
3. Sudip Misra:Introduction to Industrial Internet of Things and Industry 4.0, 2021.

Web References:

1. <https://www.tutorialspoint.com/arduino/index.htm> [Arduino Fundamentals]
2. <https://www.arduino.cc/en/Guide/> [Learning Arduino]
3. <https://funduino.de/> [Arduino Fundamentals]
4. https://www.tutorialspoint.com/internet_of_things/internet_of_things_tutorial.pdf [IoT Tutorial]
5. <https://www.guru99.com/iot-tutorial.html> [IoT Tutorial for beginners]
6. <https://www.javatpoint.com/iot-internet-of-things> [IoT Tutorial]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand fundamentals of Internet of Things System, Things and Clouds.
CO2 :	To study Microcontroller Programming for Internet of Things System.
CO3 :	Interfacing various Input / output devices in Internet of Things System
CO4 :	Interfacing various sensors and Actuator devices in Internet of Things System.
CO5 :	To develop mobile app and webpage for Internet of Things System.

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Internet of Things	√				
2	Things, clouds and Connections	√				
3	Programming Internet of Things System		√			
4	Interfacing Microcontroller with Hardware			√		
5	Sensors and Actuators for Internet of Things System				√	
6	Designing Mobile app and Webpage for Internet of Things System					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	3	3	2	-	3	2	1	-	2	2	-	3	2
CO2	1	3	3	3	-	3	2	1	-	2	2	-	3	2
CO3	2	3	3	3	-	3	2	1	-	2	2	-	3	3
CO4	1	3	3	3	-	3	2	1	-	2	2	-	3	3
CO5	1	3	3	3	-	3	2	1	-	2	2	-	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA520: Fundamental of Multimedia

(100 Marks)

Contact Hours: 03

Pre-requisite: Basic office automation skills

Methodology & Pedagogy: This course is focuses on providing hands-on experience in designing and development of Multimedia components using text, images and videos with the emphasis on job profile of Digital Marketer. Students will understand color, positioning, and other aspects of usage of Multimedia on various social media.

Outline of the course:

Week No	Practical	Description
1 - 2	Concept of colors, sizes, social media platforms	Students will learn about color combinations and some important tools/extensions, popular sizes used in various social media platforms.
3 - 4	Designing social media posts	Students will learn about designing social media post based on “Hook, Story and Offer”
5 - 6	Designing social media posts – carousal	Students will learn about designing social media post for carousal for more attention.
7 - 8	Understanding Video creation and edit	Students will learn about timeline, animation, transition, and editing of video
9 - 10	Creation of Portfolios	Students will learn to create portfolio for showcasing their creations when they start doing their own agencies.
11 - 12	Project based on learning	Students will do a small project based on whatever they have learned up to this.

Total Hours (Practical): 36

Total Hours: 36

Core Books:

1. Tay Vaughan: Multimedia, making it work: Edition 9th: Tata McGraw Hill Publication: 2014.
2. Ze-Nian Li, Mark S. Drew: Fundamentals of Multimedia : PHI: 2005

Reference Books:

1. Russell Brunson: Traffic Secrets: The Underground Playbook for Filling Your Websites and Funnels with Your Dream Customers: Hay House Business: 2020

Web References:

1. <https://designschool.canva.com/tutorials/> [Tutorials – Canva]
2. <https://help.invideo.io/> [Help – InVideo]
3. <https://www.youtube.com/c/canva> [Official YouTube channel - Canva]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be able to identify the basic concepts of multimedia and the web.
CO2 :	Be able to use various multimedia tools.
CO3 :	Be able to design and create social media posts using popular tools for various requirements.
CO4 :	Be able to design and create videos using popular tools for various requirements.
CO5 :	Apply end-user design skills of different requirements in creative industry of digital marketing and able to start own digital marketing agency.

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1-2	Concept of colors, sizes, social media platforms	√				√
3-4	Designing social media posts			√		√
5-6	Designing social media posts – carousal			√		√
7-8	Understanding Video creation and edit				√	√
9-10	Creation of Portfolios		√	√	√	√
11-12	Project based on learning	√	√	√	√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	3	2	2	1	2	2	2	2	2	1
CO2	3	1	1	3	1	3	2	1	1	2	2	2	2
CO3	2	3	3	2	1	2	2	2	2	2	3	2	3
CO4	2	3	3	2	1	2	2	2	2	2	3	2	3
CO5	2	3	3	3	2	3	3	2	2	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA521: Fundamentals of Visual Programming

(200 Marks)

Contact Hours: 06

Pre-requisite: CA 413 – Introduction to Programming & CA 419 – Introduction to Object Oriented Programming

Methodology & Pedagogy: The teacher will emphasize on explaining various concepts related to .NET framework and principles of visual programming with focus on the windows environment and c# language. In the lab session, students will be challenged to develop applications demonstrating various principles discussed during the lectures.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to .NET framework	06	36
2	Developing console applications with C#	07	
3	Developing visual applications with C#	07	
4	Developing database centric applications with C#	07	
5	Building WPF applications	05	
6	Advanced concepts in C#	04	

Total Hours (Theory): 36

Total Hours (Lab):36

Total Hours: 72

Detail Syllabus:

Unit I: Introduction to .NET framework

Hours 06

Introducing .NET Framework: .NET Framework Component, Managed Code, Common Language Runtime, Common Type System, Just-in-Time compilation process, types of applications which can be developed using .NET framework.

Unit II: Developing console applications with C#

Hours 07

Fundamentals of C#: Data types, variable declaration, type casting, conditional constructs, looping constructs, arrays, functions.

Unit III: Developing visual applications with C#

Hours 07

Life cycle of form, concept of event and event handling, working with message box, working with configuration files, designing form and working with various controls.

Unit IV: Developing database centric applications with C#

Hours 07

Introduction to ADO.NET, comparison between ADO and ADO.NET, ADO.NET architecture, concept of connected and disconnected data access, binding controls using ADO.NET objects.

Unit V: Building WPF applications

Hours 05

Overview of WPF, Features of WPF, Windows Forms vs WPF, Introduction to XAML, WPF Layout: Grid Panel, Stack Panel, Dock Panel, Wrap Panel and Canvas Panel, WPF Controls, 2D Animation in WPF.

Unit VI: Advanced concepts in C#

Hours 04

Concepts of: properties, anonymous methods, collections and multi-threading

Core Books:

1. Christian Nagel, Bill Evjen, Jay Glynn, Karli Watson, Morgan Skinner: Professional C# 2019 and .NET 4.5, Wiley – India, WROX, 2019.
2. Kogent Learning Solutions: C# Programming Black Book, Dreamtech Publication, 2018
3. Herbert Schildt: C# 4.0 the Complete Reference, McGraw Hill Publication 2019.

Reference Books:

1. E. Balagurusamy: Programming in C# a primer, McGraw Hill Publication, 2018
2. John Paul Mueller, Bill Sempf, Chuck Sphar: C# 7.0 All-in-One for Dummies, Wiley, 2018
3. Joydip Kanjilal: Mastering C# 8.0, BPB publication, 2019

Web References:

1. [http:// docs.microsoft.com/en-us/dotnet/csharp/](http://docs.microsoft.com/en-us/dotnet/csharp/) [Official C# documentation]
2. <https://visualstudio.microsoft.com/vs/> [Software download]
3. <https://www.tutorialspoint.com/csharp/index.htm> [For advanced concepts]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will be able to understand role of .NET framework as platform for application development
CO2 :	Students will be able to learn to develop console and windows applications
CO3 :	Students will be able to integrated database components with console and windows based applications.
CO4 :	Students will be able to learn user interface components from WPF and integrate with applications.
CO5 :	Students will be able to understand and improve applications with advanced features offered by C#.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to .NET framework	√				
2	Developing console applications with C#		√			
3	Developing visual applications with C#		√			
4	Developing database centric applications with C#			√		
5	Building WPF applications				√	
6	Advanced concepts in C#					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	-	-	3	1	3	-	-	-	1	1	2	1
CO2	3	2	2	3	3	3	2	-	2	2	2	3	2
CO3	3	3	2	3	3	3	3	-	2	3	2	3	1
CO4	3	3	3	3	3	3	3	2	3	3	2	2	2
CO5	3	3	3	3	3	3	3	2	3	3	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA522: Multi-paradigm programming language (100 Marks)

Contact Hours: 3

Pre-requisite: CA402: Introduction to Programming in C, CA407: Advanced Programming in C, CA503: Object Oriented Programming using C++.

Methodology & Pedagogy: The practical sessions will cover the Python Programming concepts and practices that support application development using Python Programming. The practical sessions will cover Object Oriented programming using Python Programming .

Outline of the course:

Week No	Practical	Description
1 - 2	Downloading and Installation: Installation of Python and Python documentation.	Student will be familiarize with downloading and installing Python.
3 - 4	Primary and secondary prompts, basic statements and expressions, Datatypes and Operators	Students will understand executing programs using Python. They also learn various datatypes, operators and expressions supported in Python.
5 - 6	Conditional and iterative statements	Students will to learn conditional and looping statements in Python.
7 - 8	Functions in Python: Built-in and user-defined, Magic methods	Student will be familiarize with built-in functions as well as the concept of user defined function in Python.
9 - 10	Introduction to object oriented programming	Student will be familiarize with the concept of Object Oriented Programming using Python.
11 - 12	Inheritance, Exception handling and advance Magic method.	Student will be able to understand inheritance based application development and exception handling using Python.

Total Hours (Practical): 36

Total Hours:36

Core Books:

1. Wesley J. Chun: Core Python Programming, 2nd edition, Pentice Hall, 2006.
2. Magnus Lie Hetland: Beginning Python from novice to professional, 2nd edition, Apress, 2009.
3. Magnus Lie Hetland: Beginning Python from novice to professional, 2nd edition, Apress, 2009.
3. Julia Elman, Mark Lavin: Lightweight Django: Using REST, WebSockets, and Backbone, 1st edition, O'Reilly Media, Inc, 2014

Reference Books:

1. Richard L. Halterman: Learning to Program with Python, Southern Adventist University 2011.
2. Mark Lutz: Programming Python, 4th Edition, O'Reilly, 2011.
3. Steve Holden: Python Web Programming, 1st edition, 2002.

Web References:

1. <https://www.python.org/about/gettingstarted/> [For beginners of Python]
2. <http://ocw.mit.edu/courses/electrical-engineering-and-computerscience/6-189-agent/introduction-to-programming-using-python-january-iap-2008/> [Python Materials]
3. http://people.cs.aau.dk/~normark/prog303/html/notes/paradigms_themes/paradigmoverview-section.html [Overview of Programming Paradigms]
4. <https://docs.python.org> [Python Documentation]
5. <https://www.tutorialspoint.com/django/index.htm> [Overview of Django Framework]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will be able to understand scripting and the contributions of scripting languages.
CO2 :	Clear Understanding of the built-in objects of Python and object-oriented concepts
CO3 :	Understand and implementation of functional programming
CO4 :	Be exposed to exception handling and inheritance

Course Outcomes Mapping:

Week No.	Unit Name	CO1	CO2	CO3	CO4
1-2	Downloading and Installation	√			
3-4	Getting started with Python : Datatypes, Operators and expressions		√		
5-6	Conditional and iterative statements		√		
7-8	Functions in Python			√	
9-10	object oriented programming		√		
11-12	Inheritance and Exception Handling				√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	-	-	2	-	-	-	-	-	3	1
CO2	-	3	3	-	-	-	3	1	-	-	-	3	1
CO3	2	-	2	-	3	-	1	-	-	-	-	3	1
CO4	-	-	3	3	1	-	2	-	-	-	-	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA523 Database Administration (200 Marks)

Contact Hours: 07

Pre-requisite: To be able to implement a simple DBMS and design and implement information systems using DBMS technology

Methodology & Pedagogy: During the practical session student will learn how to create procedure and function. Students can understand the concept of reusability. Student will learn how to handle error using exception and using trigger they can understand how to restrict the operation at particular time or given condition.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to PL/SQL	9	36
2	PL/SQL Stored Procedure and Trigger	8	
3	Introduction to DBA	7	
4	Transaction Management & Concurrency control	8	
5	Database Recovery Management	8	
6	Data Migration	8	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detail Syllabus:

Unit I: Introduction to PL/SQL

Hours 09

Introduction to PL/SQL, Datatypes of PL/SQL, Looping Structures, Conditional Statements, Operators, Cursor, Stored Procedure, Trigger and user defined Functions.

Unit II: PL/SQL Stored Procedure and Trigger **Hours 08**

Stored Procedure, local procedure, function, package and database trigger.

Unit III: Introduction to DBA **Hours 07**

Types of Oracle Database Users, DBA Roles and Responsibilities; Database Architecture;

ORACLE logical and physical database structure; Memory and Process Structure, SQLPLUS Overview, creating a database.

Unit IV: Transaction Management & Concurrency control **Hours 08**

Introduction to transaction, property of transactions, concurrency control concept, serializability, concurrency control problems, and its solutions- Locking techniques and two phase locking protocol.

Unit V: Database Recovery Management **Hours 08**

Failure in database system, Recovery Techniques-Check pointing, Recovery from system crash., log based recovery methods, shadow paging.

Unit VI: Data Migration **Hours 08**

About Importing, Exporting, Loading, and Unloading Data, Choosing the Right Import / Export/ Load/ Unload Option, Unloading and Loading Data, Exporting and Importing Data.

Core Books:

1. Ramez Elmasri, Shamkant B. Navathe: Fundamentals of Database Systems, 5th Edition, Pearson Publication
2. Oracle® Database Database Administrator's Guide by Mark Doran, Padmaja Potineni, Rajesh Bhatiya, Oracle Press.

Reference Books:

1. Abraham Silberschatz, Henry F. Korth, S. Sudarshan: Database System Concepts, 6th edition, McGraw Hill Publication. i Chellappan, "Big Data Analytics" Wiley 2015.
2. Raghu Ramakrishnan, Johannes Gehrke: Database Management Systems, McGraw Hill Publication
3. Anjali Jivani and Amisha Shingala- Practice book on SQL and PL/SQL with solutions, 3rd edition, Roopal book prakasan.

Web References:

1. https://docs.oracle.com/cd/B25329_01/doc/admin.102/b25107/impexp.htm [Importing, Exporting, Loading, and Unloading Data]
2. A Practical Guide to Oracle Database Administration for DBA & Developers (oracletutorial.com)
3. <https://www.tutorialspoint.com/plsql/index.htm>.

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be familiar with PL/SQL programming
CO2 :	Controlling user security
CO3 :	Define and devise transaction management, concurrency control, crash recovery components
CO4 :	Be familiar with Distributed database and Distributed Transaction management
CO5 :	Managing and controlling database network environment

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to PL/SQL	√				
2	User Access and Security		√			
3	Managing Users and Security		√			
4	Transaction Management and Concurrency control			√		√
5	Distributed database and Distributed Transaction Management				√	√
6	Importing, Exporting, Loading, and Unloading Data					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	1	2	-	1	2	2	-	-		3	3
CO2	3	3	3	3	3	3	3	3	3	3	3		3	3
CO3	2	2	2	3	1	1	2	1	2	-	-		3	1
CO4	3	3	3	3	3	3	3	3	3	2	3		3	2
CO5	3	3	2	3	3	3	2	3	3	2	3		3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA524: Data Communication and Network

(100 Marks)

Contact Hours: 04

Pre-requisite: Basic knowledge on the computer and its working.

Methodology & Pedagogy: During the theory lectures, the students will learn about role of computer networks and understand underlying concepts of digital communication. Further, the teacher will explain need for reference models and discuss detailed working of important components of reference models.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours
		Theory
1	Introduction to computer networks	09
2	Digital communication concepts	07
3	Network reference model	08
4	Working of data link layer	08
5	Working of network layer	08
6	Working of transport layer	08

Total Hours (Theory): 48

Total Hours: 48

Detailed Syllabus:

Unit I: Introduction to computer networks

Hours 09

Definition of computer network, aim and objective of computer network: applications of computer network, merits & demerits of computer network, examples of computer network, components of computer network, types of networks, network topologies, network devices: hub, switch, repeater, router, gateway, bridge.

Unit II: Digital communication concepts and Reference model

Hours 07

Parallel & serial communication, synchronous & asynchronous communication, modes of communication: simple, half-duplex, full-duplex, multiplexing, de-multiplexing: frequency division multiplexing (FDM), time division multiplexing (TDM), communication media: guided media, unguided media, switching techniques: circuit switching, message switching, packet switching.

Unit III: Network reference model

Hours 08

Need for network reference model, history of TCP/IP and OSI model, layers of OSI and TCP/IP model, comparison between TCP/IP and OSI model, Protocol Hierarchy, Network Protocols: Simple Mail Transfer Protocol (SMTP), File Transfer Protocol (FTP), Multiple Internet Mail Extension (MIME), Post Office Protocol (POP), Telnet, Domain Name Service (DNS), Dynamic Host Configuration Protocol (DHCP), Address Resolution Protocol (ARP), Reverse Address Resolution Protocol (RARP).

Unit IV: Working of Data Link Layer

Hours 08

Services provided to the network layer, framing, error control (error detection and correction code), flow control, data link layer in the internet (Serial Line Internet Protocol, Point-to-Point Protocol), mac sub layer: csma/cd/ca, IEEE standards (IEEE802.3 Ethernet IEEE 802.4 token bus, IEEE 802.5 token ring).

Unit V: Working of network layer

Hours 08

Routing Algorithms, Congestion Control Policies, Concept of Internetworking.

Unit VI: Working of Transport Layer

Hours 08

Process-to-process delivery, user datagram protocol (UDP), transmission control protocol (TCP), stream control transmission protocol (SCTP).

Core Books:

1. Andrew S. Tanenbaum: Computer networks, 4th Edition, Pearson Publication: 2019.
2. Behrouz Forouzan: Data communications and networking, 4th Edition, McGraw Hill, 2018.
3. William Stallings: Data and computer communication: 5th Edition, Pearson Publication, 2019.

Reference Books:

1. James F. Kurose, Keith W. Ross: Computer networking – A top down approach: 3rd Edition: Pearson Publication: 2017
2. Narsimha Karumanchi: Elements of computer networking: An Integrated Approach: 1st Edition: Careermonk Publication: 2014
3. Ed Tittel: Computer networking: 1st Edition, Schum's Publication: 2020

Web References:

1. www.firewall.cx/networking-topics.html [General networking concepts]
2. <https://help.sap.com/viewer/d0224eca81e249cb821f2cdf45a82ace/6.7.0.0/enUS/8c74877b86691018c9ba5f39a2fd28.html> [Transport layer concepts]
3. <https://www.plixer.com/blog/network-layers-explained/> [Network reference layers]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Student will be able to understand role of computer network in digital communication
CO2 :	Student will grasp underlying concepts related with network communication.
CO3 :	Student will understand role of reference models in standardizing communication
CO4 :	Students will learn about LAN communication and technology behind it.
CO5 :	Students will know about concepts involved in digital communication over long distance between multiple networks.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to computer networks	√				
2	Digital communication concepts		√			
3	Network reference model			√		
4	Working of data link layer				√	
5	Working of network layer				√	
6	Working of transport layer					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	-	1	1	-	-	1	1	-	3	-
CO2	3	3	3	2	1	1	-	-	-	1	-	3	1
CO3	2	2	2	2	1	2	2	1	-	1	1	2	1
CO4	3	3	3	2	1	1	2	1	-	1	2	1	1
CO5	3	3	3	2	1	2	3	1	-	2	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS111.02 C: HUMAN VALUES & PROFESSIONAL ETHICS

(100 Marks)

Contact Hours: 03

Pre-requisite courses: Ethical Leadership through Giving Voice to Values

(<https://www.coursera.org/learn/uva-darden-giving-voice-to-values?skipBrowseRedirect=true>)

Outline of the Course:

Unit. No.	Title of the unit	Minimum number of hours
1.	Introduction to Values and Ethics	05
2.	Elements and Principles of Values	08
3.	Applied Ethics	08
4.	Value, Ethics & Global Issues	05
5.	Contemporary Issues in Values and Ethics	04

Total Hours (Practical): 30
Total Hours: 30

Detailed Syllabus:

Unit – I Introduction to Values and Ethics

Hours:
05

Need, Relevance and Significance of Values General, Concept and Meaning of Values and Ethics

Unit – II Elements and Principles of Values

Hours:
08

Universal & Personal Values, Social, Civic & Democratic Value

Unit – III Applied Ethics

Hours:
08

Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics

Unit – IV Value, Ethics & Global Issues

Hours:
05

Cross-Cultural Issues, Role of Ethics & Values in Sustainability

Unit – V Contemporary Issues in Values and Ethics

**Hours:
04**

Case Studies, Presentations, Projects

Course Outcome (COs): At the end of the course, the students will be able to

CO1	Understand the concepts and mechanics of values and ethics.
CO2	Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.
CO3	Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.
CO4	Develop their responsibility towards society.
CO5	Comprehend their own core values and adhere to those values at their workplace.
CO6	Practice Ethical Leadership.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	3	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	3	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	2	-	-	-
CO5	-	-	-	-	3	-	-	-	2	-	-	-	-
CO6	-	-	-	-	-	-	-	-	2	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Reference book:

1. Human Values and Ethics in Workplace, United Nations Settlement Program, 2006. (http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
2. Ethics for Everyone, Arthur Dorbin, 2009. (<http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf>).
3. Values and Ethics for 21st Century, BBVA. (https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)

Web material:

1. www.ethics.org

DETAILED SYLLABUS

FOR

B.Sc.(IT) PROGRAMME
(5th SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - V

Course Code	Course Title	Teaching Scheme				Examination Scheme						Total
		Contact Hours			Credit	Theory			Practical			
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA616-617	Elective-V	3	3	6	6	10	20	70	15	15	70	200
CA618	Programming with JAVA	4	3	7	7	10	20	70	15	15	70	200
CA619	Data Science	3	3	6	6	10	20	70	15	15	70	200
CA620	Object Oriented Analysis and Design	4	-	4	4	10	20	70	-	-	-	100
HS131.02 C	Communication and Soft skills	-	2	2	2				30		70	100
		14	11	25	25	400			400			800

Elective-V	
Course Code	Course Title
CA616	Advanced Internet of Things
CA617	Introduction to Game Development

CA616- Advanced Internet of Things

(Marks 200)

Contact Hours: 06

Pre- requisite: CA519: Introduction to Internet of Things

Methodology & Pedagogy: In order to achieve the course objectives, students will be introduced to advanced concepts of IOT system and its structure. Various microcontrollers like node MCU, Raspberry PI and ESP32 which can be used to design IOT system. Introduction to IOT system design with specifications, development and implementation will be explained. Students will also be introduced to advance IOT programming..

Outline of the Course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	IOT and its Applications	06	36
2	IOT Architecture and IOT Protocols	06	
3	Cloud Platforms for IOT	04	
4	Embedded C and Python Programming for IOT Applications	08	
5	ESP8266 and ESP32 Microcontroller	06	
6	Raspberry PI for IOT Projects	06	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit I : IOT and its Applications

Hours: 06

What is the IoT and why is it important? Elements of an IoT ecosystem, Technology drivers, Business drivers, Trends and implications, Overview of Governance, Privacy and Security Issues. IoT applications for industry: Future Factory Concepts, Brownfield IoT, Smart Objects, Smart Applications. Study of existing IoT platforms /middleware, IoT- A, Hydra etc.

Unit II : IOT Architecture and IOT Protocols**Hours: 06**

IoT Open source architecture (OIC)- OIC Architecture & Design principles- IoT Devices and deployment models, Protocol Standardization for IoT, M2M and WSN Protocols, SCADA and RFID Protocols, Issues with IoT Standardization, Unified Data Standards Protocols – IEEE802. Zigbee, Network layer, APS layer, Security.

Unit III : Cloud Platforms for IOT:**Hours: 07**

Virtualization concepts and Cloud Architecture, Cloud computing, benefits, Cloud services, SaaS, PaaS, IaaS, Cloud providers & offerings, Study of IOT Cloud platforms, ThingSpeak API and MQTT

Unit IV : Embedded C and Python Programming for IOT Applications**Hours: 08**

Variables and Numbers, Looping Structures, Conditional Statements, Lists, Tuples and Dictionaries, Type Conversions, Function declaration, calling functions and passing values, Function Returning values. Exiting from functions, Reading/Writing to File.

Unit V : ESP8266 and ESP32 Microcontroller**Hours : 06**

Basics of Wireless Networking, Introduction to ESP8266 Wi-Fi Module, Various Wi-Fi library, Web server- introduction, installation, configuration, Posting sensor(s) data to web server, Introduction to ESP32 microcontroller.

Unit VI : Raspberry Pi for IOT Projects**Hours: 06**

Introduction to Raspberry Pi, Why Raspberry pi, Features of Raspberry pi, Basic set up and first boot configuration, Different uses of Raspberry pi, Different Versions of Raspberry pi, Vision of Raspberry pi, Different components of the Board.

Core Books:

1. Arshdeep Bahga, Vijay Madisetti: Internet of Things: A Hands-On Approach, VPT Publication, 2014.
2. Dr. Simon Monk: Programming the Raspberry Pi: Getting Started with Python, McGraw Hill Publication

Reference Books:

1. M. Richardson, S. Wallace: Getting Started with Raspberry Pi, O'Reilly, 2012
2. Dr. Simon Monk: Programming the Raspberry Pi: Getting Started with Python, McGraw Hill Publication

Web References:

1. <https://www.tutorialspoint.com/arduino/index.htm> [Arduino Fundamentals]
2. www.tinkercad.com [Arduino Simulator]
3. <https://www.youtube.com/watch?v=UUOChoCbty8> [Raspberry Pi - How to start programming with Python]
4. <https://www.raspberrypi.org/documentation/usage/python/> [Step by Step Programming]

Course Outcomes: Upon successful completion of the course, the students will:

CO1 :	Foster Ability to understand IOT structure
CO2 :	Understanding IOT Architecture
CO3 :	Sending and receiving data from cloud.
CO4 :	To study basics of Raspberry PI and Python Programming
CO5 :	To design IOT projects using ESP8266 and ESP32

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	IOT and its Applications	√				
2	IOT Architecture and IOT Protocols		√			
3	Cloud Platforms for IOT			√		
4	Embedded C and Python Programming for IOT Applications				√	
5	ESP8266 and ESP32 Microcontroller					√
6	Raspberry PI for IOT Projects					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	1	1	1	-	1	1	-	1	-	-	-
CO2	2	3	1	2	1	1	-	2	1	-	1	-	-	-
CO3	2	2	2	2	1	1	-	2	1	-	2	-	1	-
CO4	1	1	2	3	2	1	2	2	2	-	2	-	1	-
CO5	1	1	1	3	2	1	2	1	2	1	1	1	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA617: Introduction to Game Development

(200 Marks)

Contact Hours: 06

Pre-requisite: HTML, JavaScript, JQuery, CSS

Methodology & Pedagogy: : During theory lectures illustrations emphasizing the need for basic features of Games, HTML and about game development. During Practical sessions, students will be required to develop Basic games using HTML 5 Canvas and CSS.

Outline of the Course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Game Development	04	36
2	Game Design using HTML5 Canvas	07	
3	Working with Scalable Vector Graphics	06	
4	Workspace for Phaser Project Setup	05	
5	Phaser Principles and Game examples	07	
6	Basic Game Deployments	07	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit I: Introduction to Game Development

Hours 04

Introduction, Web Technologies in games, Advancements, where games are played, Web Development in Modern day, Growth of Interactive capabilities of a Web Browser, Wide selections of HTML5 Game engines, Introduction to Phaser.js, Basic Structure of an HTML5 Game project.

Unit II: Game Design using HTML5 Canvas

Hours 07

History; What Is a Canvas?; Canvas Coordinates; Registering the Canvas dimensions; Drawing on Canvas with paths, curves etc.; Working with Solid colors, Gradients & Transparency; Importing External Images & Setting the background; Working with Color & Geometrical transformations; Creating graphs & charts; Working with Text; Animating a Vertical Bar-Chart; Fine tune animation with Acceleration & Easing.; Working with Pixel Data; CSS and Canvas; Create High-Res, Retina-Display-Ready Media with Canvas; Clipping Canvas drawings & saving them to a file.; When Not to Use Canvas; Fallback Content; Implementing Canvas Security; Ensuring backward compatibility Support of Canvas API to older versions of browsers.

Unit III: Working with Scalable Vector Graphics

Hours 06

Overview of SVG, Understanding SVG, Scalable Graphics, creating 2D Graphics with SVG, Adding SVG to a Page, Simple Shapes, Transforming SVG Elements, Reusing Content, Patterns and Gradients, SVG Paths, Using SVG Text, Putting the Scene Together, building an Interactive Application with SVG, Adding the CSS Styles, Ensuring backward compatibility Support of SVG API to older versions of browsers. Integrating Video with Canvas & SVG, Applying Visual filters using Canvas & SVG.

Unit IV: Workspace for Phaser Project Setup

Hours 05

Installing Git (Node.js) using Command Prompt, CMD Navigation, Node Package Manager, Installing Bower, Installing Browserify, Babel and Babelify, setting up Basic Project, Installing Grunt Packages, writing grunt files, testing files and installing Yeoman, setting up HTML (App.js), Testing a Setup with simple game.

Unit V: Phaser Principles and Game examples

Hours 07

Game Loop, States, Display List, Camera, Loading and the Asset Cache, Images, Sprites, Sprites Sheet, Texture Atlases, Tile Atlases, Input, Sound, Maps, Tweens, Writing Tweens, Physics Primer, Phases of Physics System, Bodies, Arcade Physics, Setting up P2 world, Particles, Examples: Shoot'em Up, Game Prefabs, Tower Defense etc.

Unit VI: Basic Game Deployments

Hours 07

Web Deployment, creating a Grunt Production Task, Build Task, App Deployment, Testing, Debugging and building with CocoonJS.

Core Books:

1. Travis Faas: An Introduction to HTML5 Game Development with Phaser.js, CRC Press.
2. Andy Harris: HTML 5 Game Development for Dummies, Dummies
3. Jessica Freeman: Introducing HTML5 Game Development, O'REILLY publication

Reference Books:

1. Makzan: HTML5 Game Development HOTSHOT, Packt Publishing Limited

Web References:

1. https://www.w3schools.com/html/html5_canvas.asp

Course Outcomes: Upon successful completion of the course, the students will:

CO1 :	Be aware of Game environment
CO2 :	Be able to design Game
CO3 :	Be aware with Game graphics
CO4 :	Be capable to create interactive Games
CO5 :	Be able to deploy the game

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Game Development	√				
2	Game Design using HTML5 Canvas		√			
3	Working with Scalable Vector Graphics			√		
4	Workspace for Phaser Project Setup				√	
5	Phaser Principles and Game examples					√
6	Basic Game Deployments					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	3	-	2	1	-	-	2	-	-	-
CO2	2	-	3	-	2	1	-	-	2	2	3	2	1	1
CO3	-	3	-	2	-	1	3	2	-	-	-	3	3	2
CO4	2	1	2	2	2	1	2	-	2	-	2	2	3	2
CO5	-	2	2	3	1	-	2	2	2	2	-	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA618 Programming with JAVA

(200 Marks)

Contact Hours: 07

Objective: This course is intended to emphasize the importance of Object Oriented Programming Concepts and Java not only as an Object Oriented Programming language but as a platform for developing applications. The objectives also include extensively to train the students in Core Java Concepts. Pre- requisite: Basic knowledge of Introduction to Programming in C and OOPs concepts.

Pre-requisite: Basic Knowledge of Introduction to Programming in C and OOPs concepts.

Methodology & Pedagogy: Upon successful completion of the syllabus, students will get basic knowledge of object oriented programming. Concretely, students should be able to create appropriate classes using the Java Programming Language to solve problems using Object Oriented Approach. They should be able to write console based and multithreaded applications using the Java Programming Language.

Outline of the Course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours	
		Theory	Practical
1	Basic Introduction of Java	07	36
2	Building blocks of the Language	06	
3	Object Oriented Programming Concepts	12	
4	String Class and Exception Handling	07	
5	Packages and Multithreading	08	
6	Basics of Collections Framework	08	

Total Hours (Theory):

48

**Total Hours
(Lab): 36**

Total: 84

Detailed Syllabus:

Unit – I: Basic Introduction of Java

Hours: 07

Basics of Java, History of Java, Advantages of Java, Java Virtual Machine & Byte Code, Java Environment Setup, Java Program Structure, Difference between POP and OOP, Basics of OOP: Class, Objects, Abstraction, Inheritance, Encapsulation, Polymorphism, Create and deploy simple programs using java.

Unit – II: Building blocks of the Language

Hours: 06

Data types, Variables, Literals, Identifiers, Operators, Decision & Control flow Statements.

Unit – III: Object Oriented Programming Concepts

Hours :12

Defining classes, variables and methods, creating objects, accessing rules, this keyword, static keyword, method overloading, final keyword, Constructors: Default constructors, Parameterized constructors, passing object as a parameter, constructor overloading, Basics of Inheritance, Types of inheritance, super keyword, concepts of method overriding, extending class, dynamic method dispatch & Object class, abstract class, Interface.

Unit – IV: String Class and Exception Handling

Hours :07

String class and its methods, StringBuffer class, StringBuilder class and their operations. Exception Handling: types of exceptions, Throwable class, keywords – try, catch, throw, throws and finally, nested try statement, java built in exceptions, user defined exceptions.

Unit – V: Packages and Multithreading

Hours: 08

Creating package, importing package, access rules for packages, class hiding rules in a package. Exploring java. lang package, Wrapper classes, Boxing and Unboxing conversions. The Java Thread Model, creating a thread, Creating Multiple Threads, Thread class and the Runnable Interface, Priorities, synchronization, Inter thread communication.

Unit – VI: Basics of Collections Framework

Hours :08

The Collections Framework Overview, Collection Interfaces- Collections, List, Set, Sorted Set, Deque, The Collection classes- Array List, Linked List, Vector, Hash Set, Tree Set, Priority Queue. Accessing a collection using an Iterator, ListIterator and Enumeration, Map Interfaces and Classes: Map, Map Entry, SortedMap, and NavigableMap, HashMap, LinkedHashMap, TreeMap, Comparable & Comparator Interface.

Core Books:

1. Herbert Schildt: The Complete Reference Java J2SE, 5th Edition, TMH Publishing Company Ltd.
2. Sachin Malhotra, Saurabh Chodhary: Programming in Java, 2nd Edition, OXFORD press.

Reference Books:

1. Pravin Jain: The class of JAVA, 3rd Impression, Pearson.
2. Ivor Horton: Beginning Java 2nd JDK, 5th Edition, Wiley Computer Publishing 2007.
3. Ken Arnold, James Gosling and David Holmes: The Java Programming Language, 4th Edition, Addison Wesley.

Web References:

1. <http://www.tutorialspoint.com/java/> [For all JAVA Concepts]
2. <http://www.roseindia.net/java/> [For JAVA Tutorials with Example]
3. <https://www.javatpoint.com/java-tutorial/> [For all JAVA Concepts]

Course Outcomes: Upon successful completion of the course, the students will:

CO1 :	To introduce the object oriented programming concepts & basic building blocks of language
CO2 :	To understand basic syntax and semantics of Java, structure of java.
CO3 :	To introduce the principles of inheritance and polymorphism; and demonstrate how they relate to the design of abstract classes and Interfaces.
CO4 :	To introduce the implementation of Strings, Exception and packages
CO5 :	To introduce the concepts of multithreading & Collections.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Basic Introduction of Java	√				
2	Building blocks of the Language	√	√			
3	Object Oriented Programming Concepts			√		
4	String Class, Exception Handling & Packages			√	√	
5	Multithreading & Collections Framework				√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	2	2	1	-	-	-	1	1	3	3
CO2	3	3	3	3	3	3	-	-	-	1	3	3	3
CO3	2	2	3	3	2	2	-	-	-	2	1	2	2
CO4	3	3	3	3	3	3	-	-	-	2	3	2	2
CO5	3	3	3	3	3	3	-	-	-	1	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA619: Data Science

(200 Marks)

Contact Hours: 06

Pre-requisite: CA415- Fundamentals of Database Management System

Methodology & Pedagogy: During theory sessions topics related to Data Science will be covered with suitable examples. During Practical Sessions, students will be required to develop small applications related to reading, cleaning, analysis and visualization. Student will also get exposure of model building using machine learning algorithms. Student will get in-depth knowledge of Data Science based real applications using case study approach.

Outline of the Course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Fundamentals of Data Science	05	36
2	Statistical Data Analysis	06	
3	Reading and Cleaning Data	06	
4	Data Visualization and EDA	06	
5	Machine Learning for Data Science	08	
6	Data Science Applications and future trends	05	

Total Hours (Theory): 36

Total Hours (Lab): 36

. Total: 72

Detailed Syllabus:

Unit-I Fundamentals of Data Science

Hours 05

Introduction to Data Science, Definition and description of Data Science, Why Learn Data Science? Data Analytics Life Cycle- Data Discovery, Preparation, Model Planning, Model Building, Communicate Results. Types of Data Analysis- Descriptive, Diagnostic, Predictive, Prescriptive. Data Scientist skills set, Pros and cons of Data Science.

Unit-II Statistical Data Analysis

Hours 06

Role of Statistics in Data Science, Kind of statistics- Descriptive and Inferential Statistics Descriptive Statistics- Measure of frequency, Measure of Central Tendency, Measure of Dispersion, Measure of Position. Overview of Inferential Statistics, Probability Theory- Random Variables, Simple Probability, Conditional Probability, Binomial Distribution, Poisson Distribution, Continuous Uniform Distribution, Implementation using Python

UNIT-III Reading and Cleaning Data

Hours 06

Data Collection, reading data using different functions, Introduction to Data Preprocessing, Data types and forms, Possible data error types, Consistent Data, Data Cleaning- Filling Missing Values, smoothing noisy data, Detecting and removing outliers, Data Integration, Data transformation- Rescaling, Normalizing data, Standardizing data Implementation Using Python.

UNIT-IV Data Visualization and EDA

Hours 06

Introduction to Data Visualization, Visual encoding, Basic plots using Python, Data Visualization libraries in Python, Introduction to Exploratory Data Analysis.

Unit-V Machine Learning for Data Science

Hours 08

Overview of Machine Learning- Supervised, Unsupervised, Semi-Supervised and Reinforcement Learning Regression Methods – Simple Linear Regression, Multiple Linear Regression. Classification Methods- KNN, Logistic Regression, Naïve Bayes, SVM. Clustering Method- k-means clustering, Fuzzy c-means, Introduction of Reinforcement Learning.

Unit VI: Data Science Applications and future trends

Hours 05

Applications- Manufacturing, Personalization, Medical Image Analysis, Internet Search, Genomics, Speech Recognition, Advertising, Financial Fraud Detection, Airline Route Planning, Cyber-threats, transportation, gaming, recommendation system.

Core Books:

1. Sanjeev J, Wagh, Manisha S. Bhende, Anuradha D. Thakare : Fundamentals of Data Science: First Edition: CRC Press : 2023
2. Dr. Gypsy Nandi, Dr.Rupamkumar Sharma: Data Science Fundamentals and Practical Approaches: BPB Publications:2020

Reference Books:

1. Parikshit Narendra Mahalle, Gitanjali Rahul Shinde, Priya Dudhale Pise, Jyoti Yogesh Deshmukh:
Foundation of Data Science for Engineering Problem Solving: Springer:2022
2. K.G.Srinivasa,G.M.Siddesh,ChetanShetty,SowmyaB.J:Statistical Programming in R:
First Edition: Oxford Press: 2017.
4. Lillian Pearson : Data Science for Dummies,John Willey & Sons:2015
5. Rachel Schutt, Carthy O'Neil: Doing Data Science: First Edition:O'Reilly: 2014

Web References:

1. <https://www.cs.cornell.edu/jeh/book.pdf> [Foundation of Data Science E-book]
2. <https://www.edureka.co/blog/math-and-statistics-for-data-science/> [Essentials Mathematics and Statistics of Data Science]
3. https://www.w3schools.com/datascience/ds_analyze_data.asp [Data Preparation using tools]
4. <https://www.kdnuggets.com/2022/06/plotting-data-visualization-data-science.html> [Data Plotting and Visualization]
5. <https://www.simplilearn.com/10-algorithms-machine-learning-engineers-need-to-know-article> [Machine Learning Algorithms]
6. <https://www.geeksforgeeks.org/major-applications-of-data-science/> [Applications of Data Science]

Course Outcomes: Upon successful completion of the course, the students will,

CO1 :	Able to understand fundamentals of Data Science.
CO2 :	Able to use statistical data analysis techniques for Data Science Applications
CO3 :	Able to apply data cleaning, visualization and exploratory data analysis techniques on data.
CO4 :	Able to build model using machine learning techniques.
CO5 :	Able to understand applications and future trends of Data Science.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Fundamentals of Data Science	√				
2	Statistical Data Analysis		√			
3	Reading and Cleaning Data			√		
4	Data Visualization and EDA			√		
5	Machine Learning for Data Science				√	
6	Data Science Applications and future trends					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA620: Object Oriented Analysis and Design

(200 Marks)

Contact Hours: 04

Pre-requisite: None

Methodology & Pedagogy: This course focuses on providing understanding of analysis and designing of large-scale software systems with emphasis on object oriented paradigm for software development. Students will be able to generate concrete software engineering artifacts at all stages of the system development life cycle. The course will enrich students with analysis and design principles and methods for large-scale software system development using object oriented methods.

Outline of the Course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Fundamentals of Information Systems	06
2	Systems Development Life Cycle and Requirements Determinations	09
3	Introduction to Object Oriented Paradigm	06
4	Object Oriented Concepts	09
5	System Design Using UML	09
6	Computer-Aided Systems Tools and Software Quality	09

Total Hours (Theory): 48

Total Hours: 48

Detailed Syllabus:

Unit I : Fundamentals of Information Systems

Hours 06

General Architecture of Systems with basic components, Open and Close Systems, TPS, MIS, DSS and ES Types of Systems, Examples of Real-life Systems, Various users of Systems.

Unit II : Systems Development Life Cycle and Requirements

Determinations

Hours 09

Phases of the Classical Systems Development Life Cycle (SDLC) Method: Preliminary Investigation, Determination of Requirements, Request Clarification, Design of System, Development of System, System Testing, Implementation, Maintenance and Support

Unit III : Introduction to Object Oriented Paradigm

Hours 06

Introduction, Object Oriented System Development Methodology, Difference between traditional approach and OO approach, Objects, Classes, Attributes, Behavior and methods

Unit IV : Object Oriented Concepts

Hours 09

Abstraction, Encapsulation, Class Hierarchy, Inheritance, Multiple Inheritance, Polymorphism, Relationships and association, Aggregation

Unit V : System Design Using UML

Hours 09

Introduction to Class diagrams, Class Diagram and Object Diagram, Activity diagram, Introduction to dynamic system design, Sequence diagrams, Collaboration diagrams

Unit VI : Computer-Aided Systems Tools and Software Quality

Hours 09

Introduction to CASE Tools, Advantages and Weakness of CASE, Software Quality Parameters, Various Types of Testing and Test Data: Live Data, Artificial Data, Test Libraries, Training Users, Objectives for Training, Documentation

Core Books:

1. James A. Senn: Analysis and Design of Information System: Second Edition: McGraw Hill International: 2010.
2. Timothy C. Lethbridge and Robert Laganière: Object-Oriented Software Engineering: Practical Software Development using UML and Java: Second Edition: McGraw-Hill Education: 2005.

Reference Books:

1. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modeling Language User Guide: Tenth Impression: Pearson Education, Inc : 2011

Web References:

1. http://www.cs.uic.edu/~jbell/CourseNotes/OO_SoftwareEngineering

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Understanding of different types and users of system, Details of SDLC phases, Concepts of feasibility study
CO2 :	Understanding of object oriented paradigm, fundamentals of Objects, Classes, Attributes, Behavior and methods
CO3 :	Gaining knowledge about object oriented concepts such as abstraction, encapsulation, polymorphism, inheritance, etc.
CO4 :	Understanding analysis and designing with various UML diagrams
CO5 :	Understanding modern tools for object oriented systems analysis and design

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Fundamentals of Information Systems	√				
2	Systems Development Life Cycle and Requirements Determinations	√				
3	Introduction to Object Oriented Paradigm		√			
4	Object Oriented Concepts			√		
5	System Design Using UML				√	
6	Computer-Aided Systems Tools and Software Quality					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	3	1	1	3	3	3	-	3	1	3	3
CO2	1	2	2	1	1	3	1	1	-	1	1	3	1
CO3	2	2	2	1	1	3	2	1	-	1	1	3	1
CO4	1	3	3	3	1	2	1	3	-	2	3	3	3
CO5	1	1	1	3	1	2	1	3	-	1	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

DETAILED SYLLABUS

FOR

B.Sc.(IT) PROGRAMME
(6th SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 – 24
Semester - VI

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA621	Basics of Mobile Applications	3	3	6	6	10	20	70	15	15	70	200
CA622	Software Testing	-	3	3	3	-	-	-	15	15	70	100
CA623	Mini Project	-	15	15	15	-	-	-	100		200	300
HS132.02 C	Contributor Personality Development	-	2	2	2	- -		-	30		70	100
		03	23	26	26	100			600			700

CA621: Basics of Mobile Applications

(200 Marks)

Contact Hours: 06

Pre-requisite: Object Oriented Programming using JAVA

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for basic features of Mobile Computing and Android- the Mobile Application Development platform will be given. During Practical sessions, students will be required to develop Mobile Application using JAVA programming language in Android. Student shall also develop applications with elegant user interface that deal with data storage, documents sharing among applications.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction Android with Kotlin	07	36
2	Android Layouts and Views	06	
3	Android Widgets and Adapters	06	
4	Advanced Designing: Material Design	05	
5	Data Storage: SQLite, Firebase	07	
6	Deploying Web Services	05	

Total Hours (Theory):36

Total Hours (Lab):36

Total Hours:72

Detail Syllabus:

Unit I: Introduction Android with Kotlin

Hours 07

Introduction to Mobile Applications and Android, Basics of Kotlin, setting up development environment, Dalvik Virtual Machine, Android Application Structure, DDMS, Android Manifest File, Gradle, Android permissions, APK, Basic Building blocks – Activities, Services, Broadcast Receivers & Content providers.

Unit II: Android Layouts and Views

Hours 06

Working with Containers: LinearLayout, RelativeLayout, TableLayout, ScrollView, and FrameLayout, Fragment, Views: Toast, TextView, EditText, Button, ImageButton, CheckBox, ToggleButton, RadioButton, RadioGroup, ProgressBar, AutoComplete TextView, Components for communication -Intents & Intent Filters Intents: Explicit Intents, Implicit Intents, switching between activities and passing data between activities using Intents, Logging.

Unit III: Android Widgets and Adapters

Hours 06

Calendar: Manage Date and Time, Adapters: ArrayAdapter, BaseAdapter ListView and ListActivity, GridView, Gallery Menus: Option menu, Context menu, Sub menu Tabs and TabActivity Alert Dialogs, Android Toolbar

Unit IV: Advanced Designing: Material Design

Hours 05

Android Design Support Library, Vector Drawables, Basics of material design components and principles, Android styles and themes, Animations and Transitions

Unit V: Data Storage: SQLite, Firebase

Hours 07

Basics of Storage in Android, Shared Preference: Save key-value data, Store Data in Database, store data in a firebase, save files on Device, Sharing of data.

Unit VI: Deploying Web Services

Hours 05

Working with Web Service in Android, Deploying Web Service

Core Books:

1. Wei-Meng Lee: Beginning Android 4 Application Development, Wiley India Pvt Ltd.
2. Reto Meier: Professional Android 2 Application Development, Wrox publication
3. Mark L. Murphy: The Busy Coder's Guide to Android Development

Reference Books:

1. Dawn Griffiths, David Griffiths: Head First Android Development, O'REILLY publication
2. Mark L. Murphy: Beginning Android 2, APRESS publication
3. Mark L. Murphy: The Busy Coder's Guide to Advanced Android Development, Commons Ware, LLC

Web References:

1. <https://kotlinlang.org/> [Kotlin Tutorials]
2. <https://developer.android.com/guide/> [Android Tutorials]
3. www.vogella.com/tutorials/android.html [Android Practical Tutorials]
4. <https://www.androidhive.info/> [Android Practical Tutorials]

Course Outcomes: Upon successful completion of the course students will be,

CO1 :	Be acquainted with Android OS
CO2 :	Be aware with Android Development Tools
CO3 :	Be able to create proper look and feel of mobile application
CO4 :	Be capable to create interactive mobile application that handle data
CO5 :	Be able to create mobile application that works web service.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction Android with Kotlin	√				
2	Android Layouts and Views		√			
3	Android Widgets and Adapters			√		
4	Advanced Designing: Material Design				√	
5	Data Storage: SQLite, Firebase				√	
6	Deploying Web Services					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	3	-	2	1	-	-	2	-	-	-
CO2	2	-	3	-	2	1	-	-	2	2	3	2	1	1
CO3	-	3	-	2	-	1	3	2	-	-	-	3	3	2
CO4	2	1	2	2	2	1	2	-	2	-	2	2	3	2
CO5	-	2	2	3	1	-	2	2	2	2	-	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA622: Software Testing

(100 Marks)

Contact Hours: 03

Pre-requisite: Object oriented programming concepts.

Methodology & Pedagogy: The practical sessions will cover the software testing concepts and practices that support the production of quality software. The practical sessions will cover the application of testing techniques at various levels of testing using manual and automated testing tools.

Outline of the course:

Week No	Practical	Description
1 - 2	Introduction to testing, Checklist preparation	Student will be able to understand role of testing in real world and he/she can prepare checklist for given problem.
3 - 4	Manual Testing concepts: Understanding of Static testing and Dynamic testing	Student will be familiarize with methods of testing which can be applied by inspecting, reviewing, or preparing walkthroughs.
5 - 6	Creating test case from use case scenarios and test scenarios.	Students will going to learn difference between test case and test scenarios. They can also prepare checklists by using UML use case scenarios.
7 - 8	Hand-on exercise on concepts of levels of testing : Unit Testing with JUnit	Student will be familiarize with different concepts of Junit and perform unit testing with using the same.
9 - 10	Testing WebApp using selenium web driver with JAVA/Python	Student will be familiarize with different concepts of Web driver and perform web application testing with using the same.
11 - 12	Understanding of basics Mobile Application testing using Appium and Demonstration of defect tracking tool such as JIRA/ Cucumber etc.	Student will be able to understating basics of mobile application testing using Appium.

Total Hours (Practical):36

Total Hours:36

Core Books:

1. Sagar Naik, Piyu Tripathy: Software Testing and Quality Assurance, Theory and Practice, Wiley, 2008.
2. Paul C. Jorgensen : Software Testing: A Craftsman's Approach, 4th Edition by , CRC press, Taylor and Francis Group, 2014
3. Roger S Pressman: Software Engineering – A Practitioner's Approach, 7th Edition, McGRAW HILL International Edition, 2010.

Reference Books:

1. Mauro Pezze, Michael Young: Software testing and Analysis- Process, Principles and Techniques, Wiley India, 2012.
2. Boris Beizer: Software Testing Techniques: 2nd Edition, Van Nostrand Reinhold, 1990.
3. Daniel Galin: Software Quality Assurance, Pearson Education, 2004.
4. Ron Patton: Software Testing, Pearson Education, 2005.

Web References:

2. <http://tryqa.com/what-is-software-testing> [Basic of Software Testing]
3. <https://www.guru99.com/functional-testing.html> [Functional Testing]
4. <http://www.softwaretestinggenius.com/download/bgstpadmini.pdf> [Software Testing Life Cycle]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to understand role, principles of software testing.
CO2 :	Familiar with the structural techniques of testing.
CO3 :	Understand the concepts of creating test cases with respect to levels of testing.
CO4 :	Learn basics of Unit testing.
CO5 :	Learn basics of web application and mobile application testing.

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1-2	Introduction to checklist preparation	√				
3-4	Creating test cases		√			
5-6	Writing test cases			√		
7-8	Familiar with Unit testing using JUnit				√	
9-10	Understanding Selenium Web driver					√
11-12	Testing mobile app using Appium					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2	2	3	2	2	3	2	2	2	3	2
CO2	3	2	2	2	3	2	2	1	1	2	1	1	1	2
CO3	3	2	2	1	2	3	2	1	1	1	2	1	1	2
CO4	3	2	1	1	2	2	2	2	1	2	1	2	1	1
CO5	3	3	2	1	2	1	2	1	2	2	1	2	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA623: Mini Project

(300 Marks)

Contact Hours: 15

Objective:

The main objective of this mini project is to let the students apply the programming knowledge to a real-world situation/problem and exposed the students with specific programming skills and help in developing a working model in terms of application.

Course Outcomes:

Co1:	Student will understand the implementation of concepts of SDLC and Software Engineering.
Co2:	The programming concepts they learn during their academics, it will be converted in to the actual implementations.
Co3:	Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.
Co4:	Students must understand the difference between a program and professional application/product/software.
Co5:	Students will learn different categories of applications like Desktop application, Web applications, etc.

Guidelines:

Mini Project is in house project development. Every student is required to carry out Mini Project work under the supervision of a guide provided by the placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Mini Project work during the semester as per the schedule provided by the placement Coordinator.

Mini Project proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Mini Project to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. Placement coordinator will provide the format of project report. Student has to submit one copy of Mini Project to the institute. Each Student is required to make a copy of Mini Project in CD and submit along with Mini Project report.

Evaluation:

The project must be evaluated in two aspects:

- a. Internal (100 Marks):
 - i.Reporting to internal project guide
 - ii.Incorporation of suggestions by project guide
 - iii.Internal Project viva examination
- b. External (200 Marks):
 - i.Project Report Preparation & Evaluation
 - ii.External Project Viva Examination

Course Code	Course Title	Teaching Scheme		Internal	End Semester Examination		Total
		Contact Hours	Credit	Continuous Evaluation	Report	Presentation & Viva	
CA623	Mini Project	15	15	100	100	100	300

Web References:

1. <http://techwhirl.com/writing-software-requirements-specifications/>[For effective SRS]
2. http://www.ibm.com/developerworks/websphere/library/techarticles/o3o6_perks/perks2.html [For best practices of Software Project Development]
3. http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf [Requirement analysis guidelines]
4. <http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf>[Software Design Principles and Guidelines]
5. <http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%20Testing%20-2050%20specific%20ways%20to%20improve%20your%20testing.pdf>[For Effective Software Testing]
6. [http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template .pdf](http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template.pdf) [For guidelines to prepare software project report]
- 7.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO2	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO3	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO4	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO5	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation

DETAILED SYLLABUS

FOR

B.Sc.(IT) PROGRAMME
(7th SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 – 24
Semester - VII

Course Code	Course Title	Teaching Scheme				Examination Scheme							
		Contact Hours			Credit	Theory			Practical			Total	
		Theory	Practical	Total		Internal		External	Internal		External		
						Case Study	Tests		Term work	Tests			
CA624	Online MOOC Course on Current Trends of Information Technology #	-	-	-	03	-	-	-	-	-	-	-	
CA625	Minor Project	-	24	24	24	-	-	-	200		400	600	
		-	24	24	27	-			600			600	

Students can select any MOOC course of their choice related to current trends of Information Technology with at least 30 hours. 03 credits will be transferred in the result of students based on this course.

CA625: Minor Project (600 Marks)

Contact Hours: 24

Objective:

The main objective of this Minor project is to let the students apply the programming knowledge to a real-world situation/problem and exposed the students with specific programming skills and help in developing a working model in terms of application.

Course Outcomes:

Co1:	Student will understand the implementation of concepts of SDLC and Software Engineering.
Co2:	The programming concepts they learn during their academics, it will be converted in to the actual implementations.
Co3:	Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.
Co4:	Students must understand the difference between a program and professional application/product/software.
Co5:	Students will learn different categories of applications like Desktop application, Web applications, etc.

Guidelines:

Minor Project is in house project development. Every student is required to carry out Minor Project work under the supervision of a guide provided by the placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Minor Project work during the semester as per the schedule provided by the placement Coordinator.

Minor Project proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Minor Project to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. Placement coordinator will provide the format of project report. Student has to submit one copy of Minor Project to the institute. Each Student is required to make a copy of Minor Project in CD and submit along with Minor Project report.

Evaluation:

The project must be evaluated in two aspects:

- a. Internal (200 Marks):
 - i.Reporting to internal project guide
 - ii.Incorporation of suggestions by project guide
 - iii.Internal Project viva examination
- b. External (400 Marks):
 - i.Project Report Preparation & Evaluation
 - ii.External Project Viva Examination

Course Code	Course Title	Teaching Scheme		Internal	End Semester Examination		Total
		Contact Hours	Credit	Continuous Evaluation	Report	Presentation & Viva	
CA625	Minor Project	24	24	200	200	200	600

Web References:

1. <http://techwhirl.com/writing-software-requirements-specifications/>
[For effective SRS]
2. http://www.ibm.com/developerworks/websphere/library/techarticles/o306_perks/parks2.html [For best practices of Software Project Development]
3. http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf [Requirement analysis guidelines]
4. <http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf>
[Software Design Principles and Guidelines]
5. <http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%20Testing%20-%2050%20specific%20ways%20to%20improve%20your%20testing.pdf>[ForEffective Software Testing]
6. [http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template .pdf](http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template.pdf)[For guidelines to prepare software project report]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO2	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO3	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO4	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO5	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation

DETAILED SYLLABUS

FOR

B.Sc.(IT) PROGRAMME
(8th SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Science in Information Technology (B.Sc. (IT)) Programme
(Choice Based Credit System)
Effective from Year 2023 – 24
Semester - VIII

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA626	Dissertation	-	24	24	24	-	-	-	200		400	600
		-	24	24	24	-			600		600	

CA626: Dissertation

(600 Marks)

Contact Hours: 24

Objective : To allow students to undertake a substantial in-depth study of a specific topic in Computer Science

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to identify and investigate a research problem in computer science
CO2 :	Able to apply an appropriate research design and associated methods rigorously
CO3 :	Able to draw appropriate conclusions and indicate the significance of findings
CO4 :	Able to report the research in scholarly fashion.

Guidelines:

- Decide the topic of dissertation with consultation of your supervisor.
- It is mandatory to report to supervisor two times in month.
- Decide the content and flow of dissertation with the consultation of your supervisor
- Before submission of dissertation report, it is expected from student to publish/present at least one research paper in National/International conference/journal of repute.
- Submit two spiral binding copies of your dissertation duly signed by your supervisor and Head of the department.
- The dissertation evaluation criteria are as follows:
 - Research topic and familiarity with the theory
 - Literature Survey
 - Research Methodologies used
 - Discussion and Conclusions
 - Structure and Presentation

- Teaching and Examination scheme of the course is as follows:

Teaching Scheme				Examination Scheme						
Contact Hours			Credit	Theory			Practical			Total
Theory	Practical	Total		Internal		External	Internal		External	
				Case Study	Test		Term work	Test		
-	24	24	24	-	-	-	200		400	600
-	24	24	24	-			600			600

Web References:

- <https://dominikryst.com/posts/cs-dissertation-tips/> [Dissertation Tips]
- <https://researchguides.njit.edu/compsci/litreview> [Literature Review Guidelines]
- <https://www.ic.unicamp.br/~wainer/cursos/2s2006/epistemico/ramesh.pdf>[Research in Computer Science]
- <https://www.indeed.com/career-advice/career-development/how-to-write-a-conclusion-for-a-research-paper> [How to write conclusions]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	1	1	3	3	2	2	2	3	2	3	3
CO2	3	3	3	2	3	3	2	2	2	3	2	3	3
CO3	2	3	1	3	3	3	2	2	2	3	2	3	3
CO4	-	1	-	2	3	3	2	2	2	3	2	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”